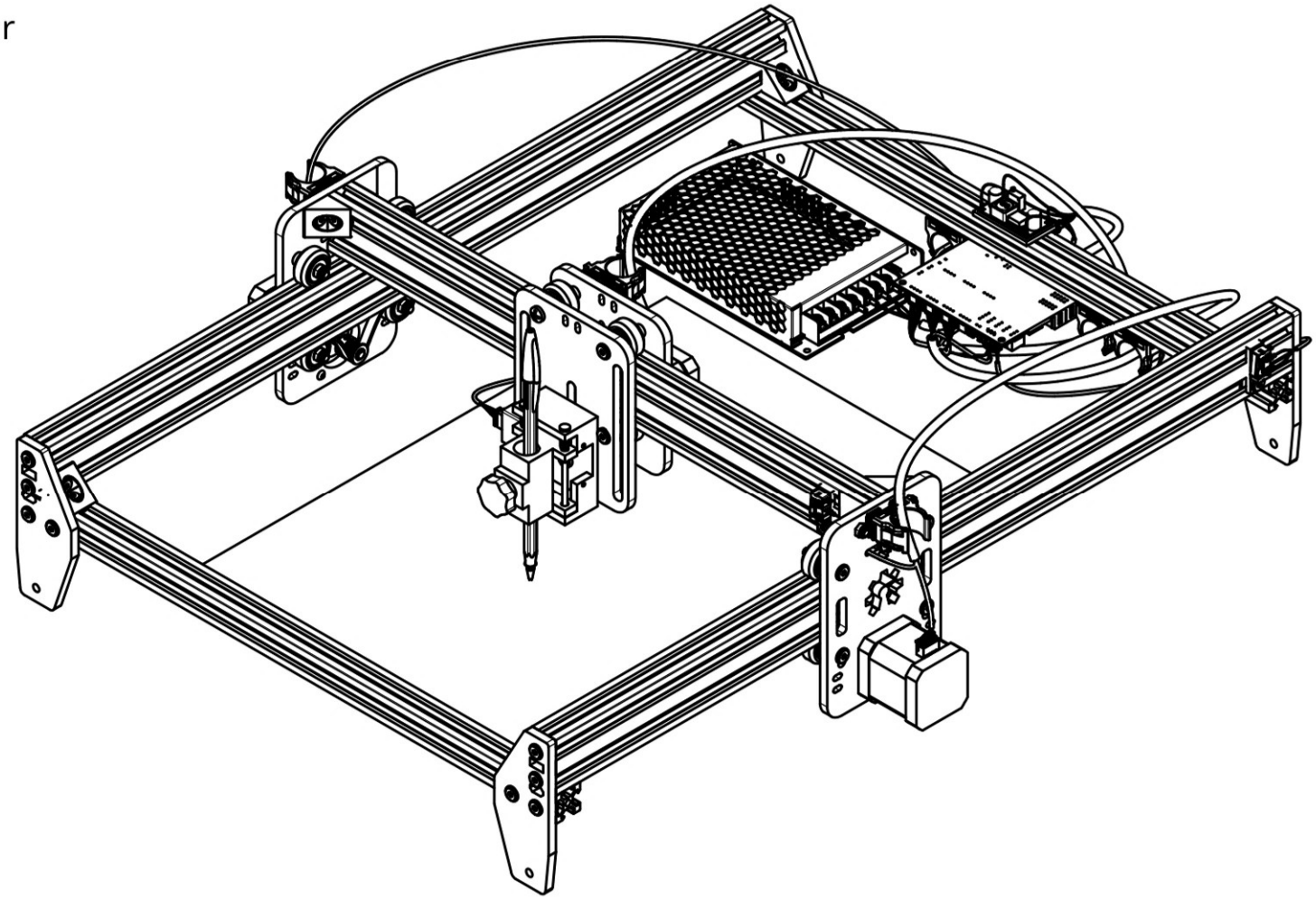
CNC Pen Ploter assembly instructions

By Nitin Swarnkar



Parts Needed

Category	Part	Specification	Qty
Motion	NEMA 17 Stepper Motor	42 × 42 mm	3
	Delrin Solid V-Wheel	—	12
	GT2 Timing Pulley	20T, M5	3
	GT2 Timing Belt	2 m	1
Aluminium Frame	V-Slot Extrusion 2040	610 mm	2
	V-Slot Extrusion 2040	444 mm	1
	V-Slot Extrusion 2020	444 mm	2
	Corner Bracket	20 Series	8
M5 Hardware	Ultra Low Profile Screw	M5x 50mm	1
	Ultra Low Profile Screw	M5 x40mm	3
	Ultra Low Profile Screw	M5x 30mm	8
	Ultra Low Profile Screw	M5 × 25 mm	12
	Ultra Low Profile Screw	M5 × 15 mm	6
	Ultra Low Profile Screw	M5 × 10 mm	3
	Ultra Low Profile Screw	M5 × 8 mm	30
	Nylon Lock Nut	M5	16
	Washer / Shim	M5	28
	Drop-In T-Nut	M5	34

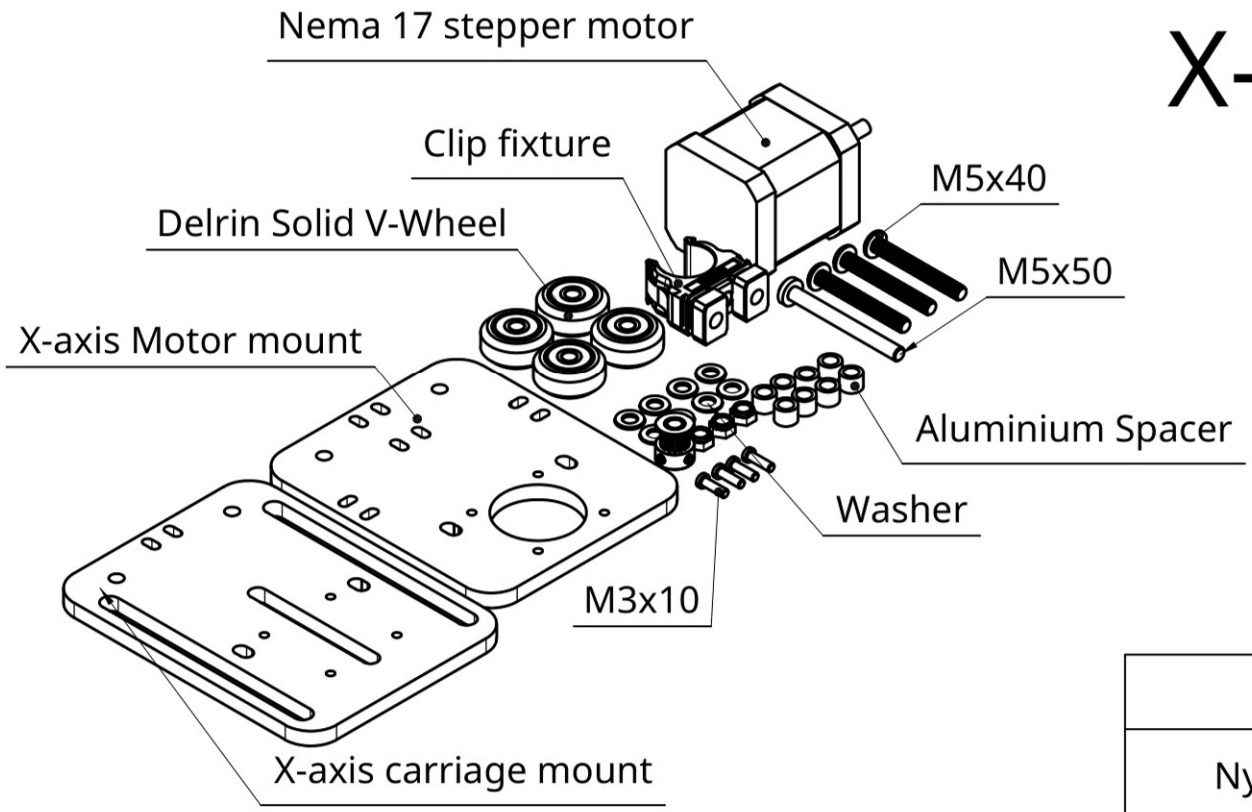
Parts Needed

Category	Part	Specification	Qty
M3 Hardware	Screw	M3 × 10 mm	12
	Screw	M3 × 16 mm	2
	PCB Stand-Off	10 mm	2
	Drop-In T-Nut	M3	2
M2.5 Hardware	Screw	M2.5 × 4 mm	2
M2 Hardware	Screw	M2 × 8 mm	6
Spacers	Aluminium Spacer	5.1 mm ID × 6 mm	16
Printed Parts	Clip Fixture	—	6
	Limit Switch Mount	—	2
	Pen Carriage Moving	—	1
	Pen Carriage Monunt	—	1
	Buck convertor Mount	—	1
Laser cut	X-Axis Motor Mount	6mm thick	1
	X-Axis Carriage Mount	6mm thick	1
	Y-Axis Mount	6mm thick	2
	Legs	6mm thick	4
Pen Lift	Concrete Nail	70 mm	2
	Pen Spring	7.5 mm	2
	Screw Knob	M5	1

Parts Needed

Category	Part	Specification	Qty
Electronics	MKS DLC32	V2.1	1
	DRV8825 Motor Driver	—	3
	Meanwell Power Supply	LRS-150-24, 24 V	1
	DC-DC Converter	XL6009 Boost-Buck	1
	Limit Switch	—	2
Connectors	JST-XH Connector Kit	2.54 mm, 230 pcs	1 kit
Tool	Wire Stripper / Crimper	27 in 1	1
wires	22 AWG wire.	RED	5 m
	22 AWG wire.	BLACK	3 m
	22 AWG wire.	BLUE	3 m
	22 AWG wire.	GREEN	3 m
	22 AWG wire.	YELLOW	1 m

X-Axis Assembly



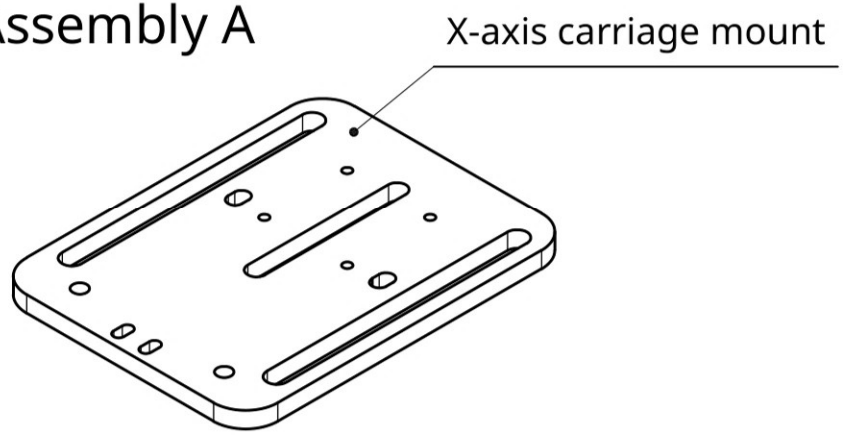
PARTS NEEDED :

Name	Size	Quantity
Clip fixture	-	1
Washer	M5 Shim	8
X-axis Motor mount	-	1
X-axis carriage mount	-	1

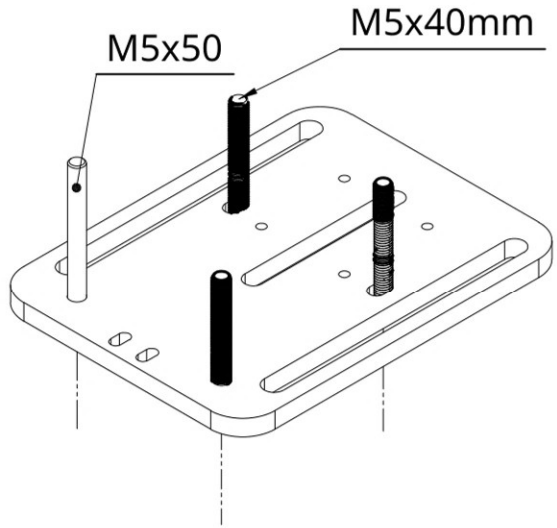
Name	Size	Quantity
Nylon Lock Nut	M5	3
40mm Ultra Low Profile Screw	M5	3
50mm Ultra Low Profile Screw	M5	1
Aluminium Spacer	5.1mm ID × 6mm	8
10mm Ultra Low Profile Screw	M3	4
Nema 17 stepper motor	42x42mm	1
Delrin Solid V-Wheel	-	4

X-Axis Cariage

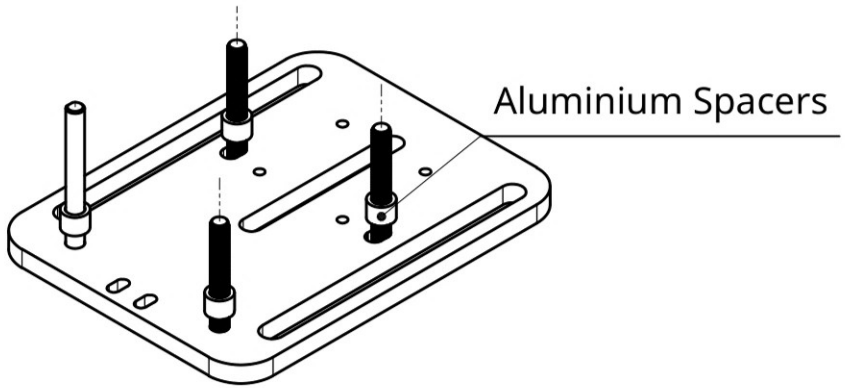
1. Assembly A



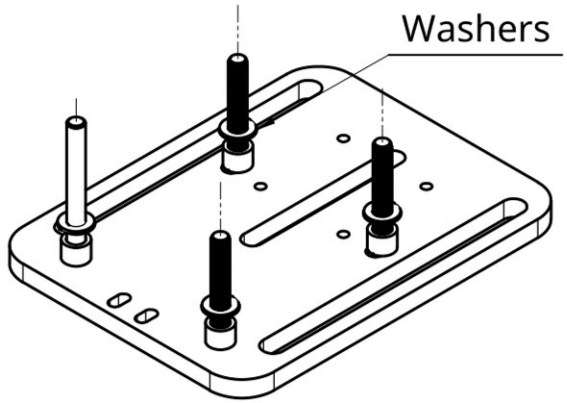
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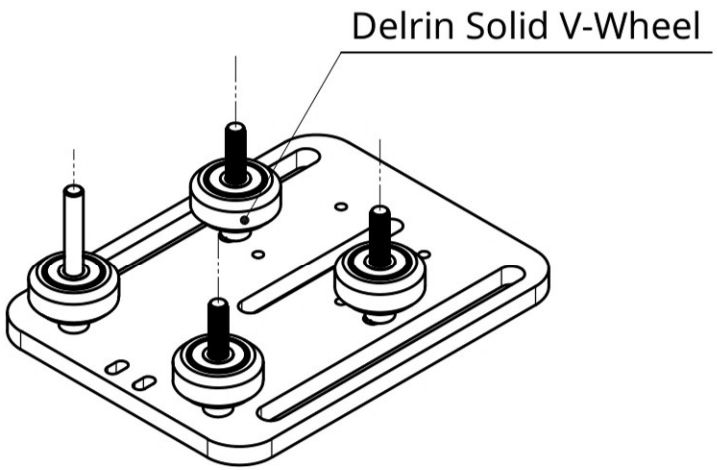
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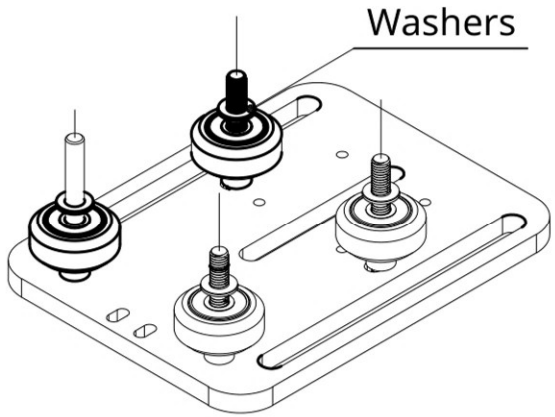
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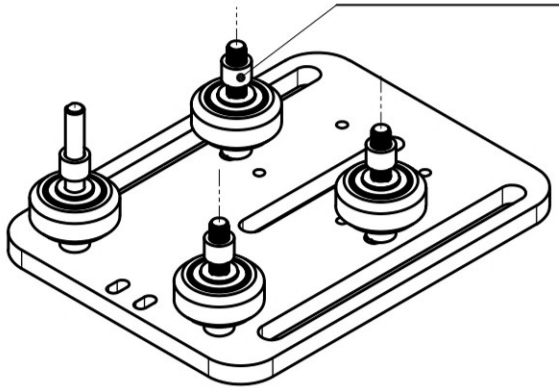
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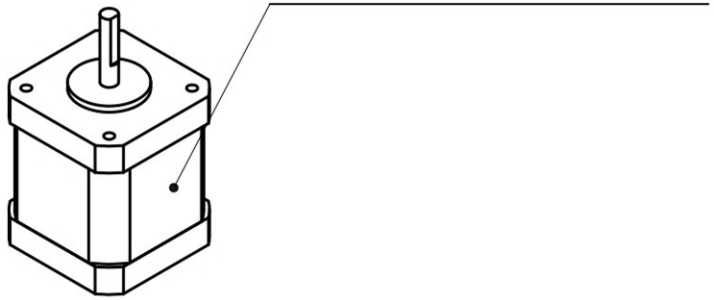
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7

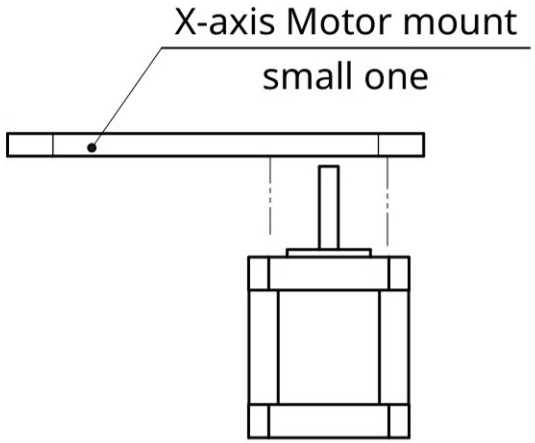


8. Assembly B

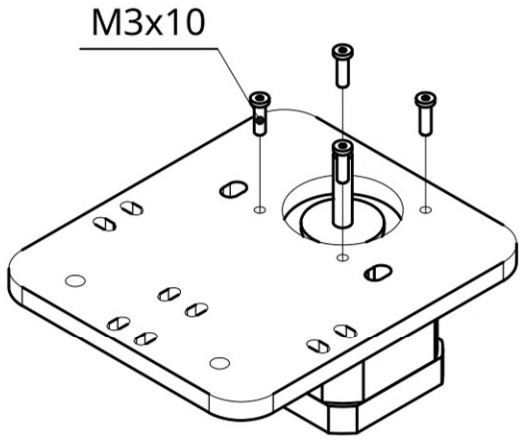


DWG NO.	SIZE	SCALE	SHEET	REV.
	A4	1:2	7 of 41	

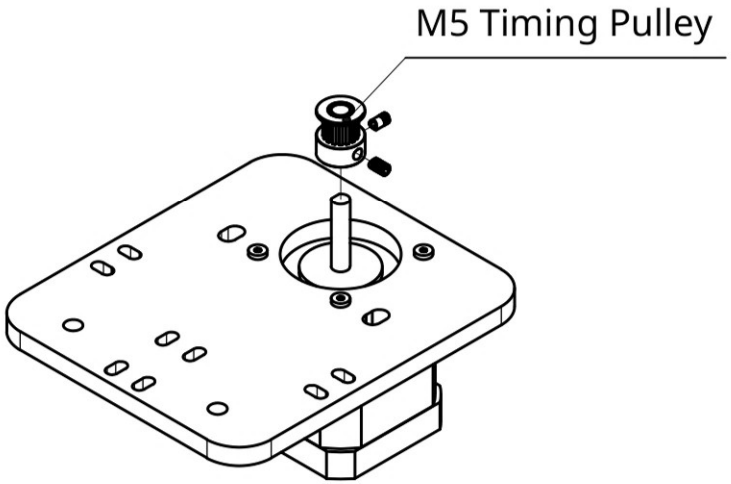
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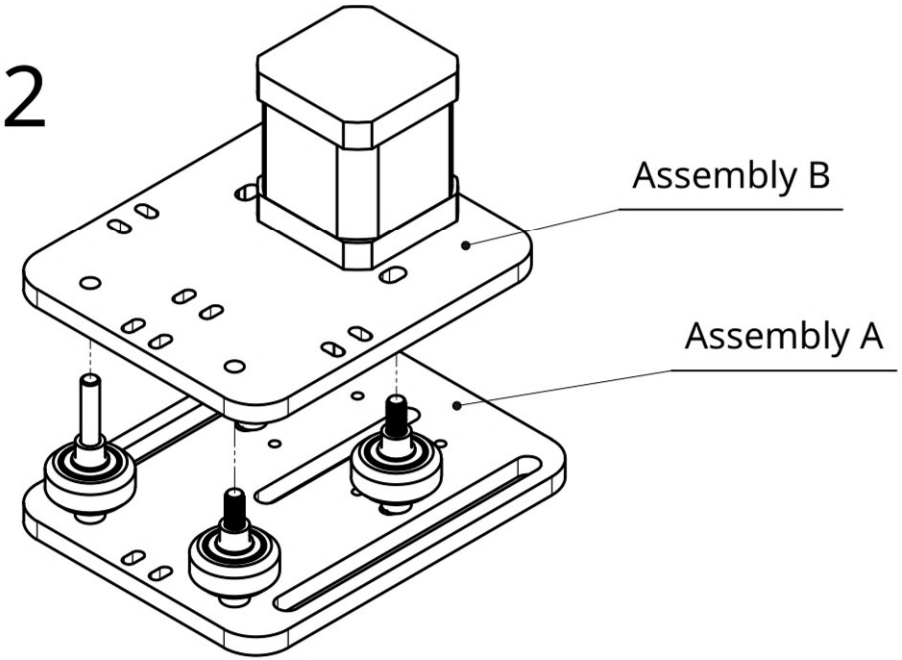
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11

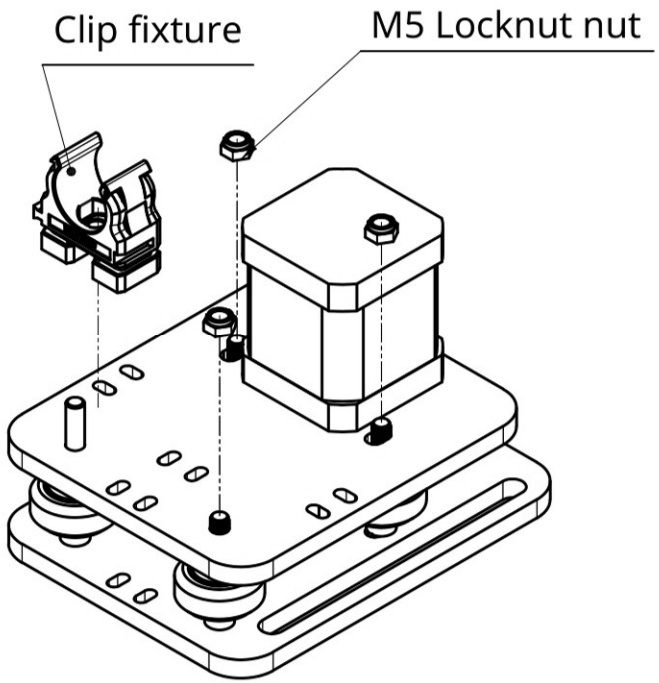


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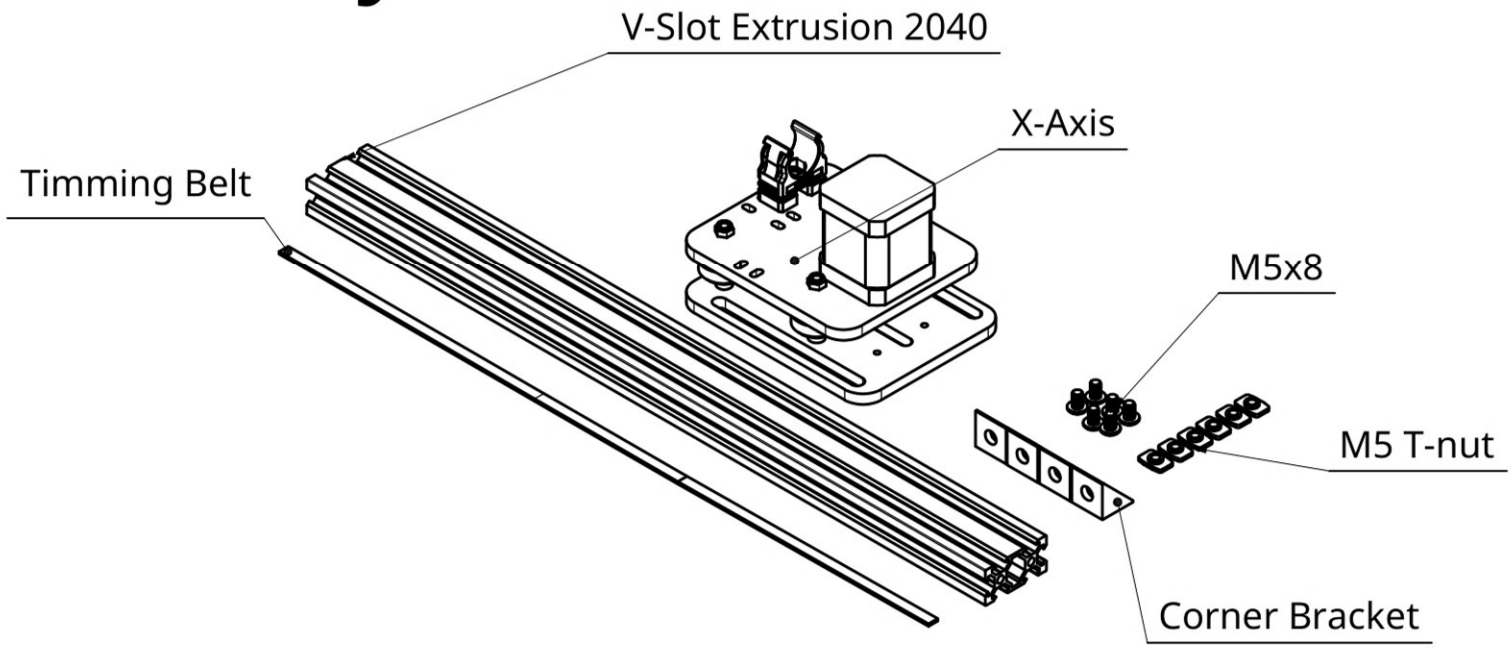
DWG NO.	SIZE	SCALE	SHEET	REV.
	A4	1:2	8 of 41	

13



DWG NO.	SIZE	SCALE	SHEET	REV.
	A4	1:2	9 of 41	

X-Axis Assembly

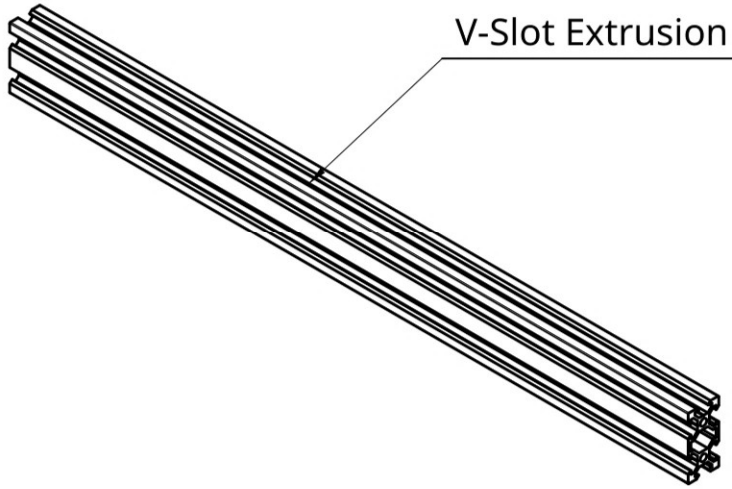


PARTS NEEDED :

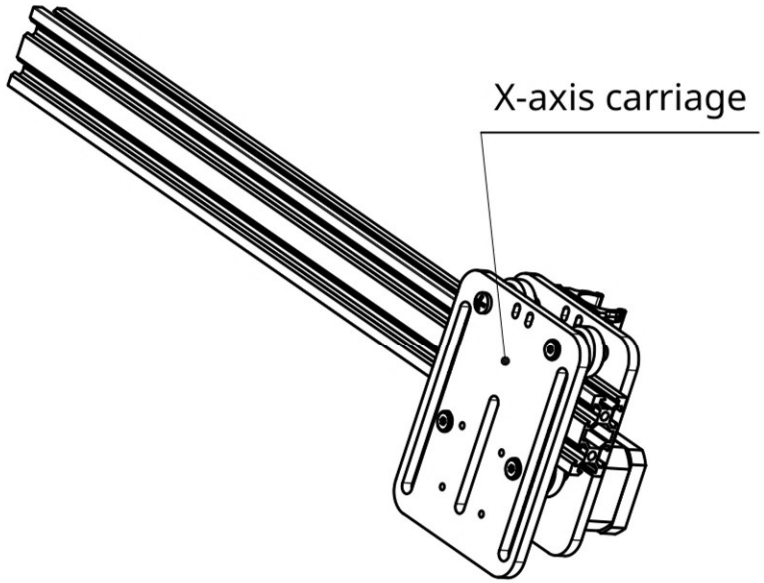
Name	Size	Quantity
8mm Ultra Low Profile Screw	M5	6
Drop-In T-Nut	M5	6
Corner Bracket	20 Series	4
V-Slot Extrusion 2040	444 mm	1
GT2 Timing Belt	450 mm	1

X-Axis

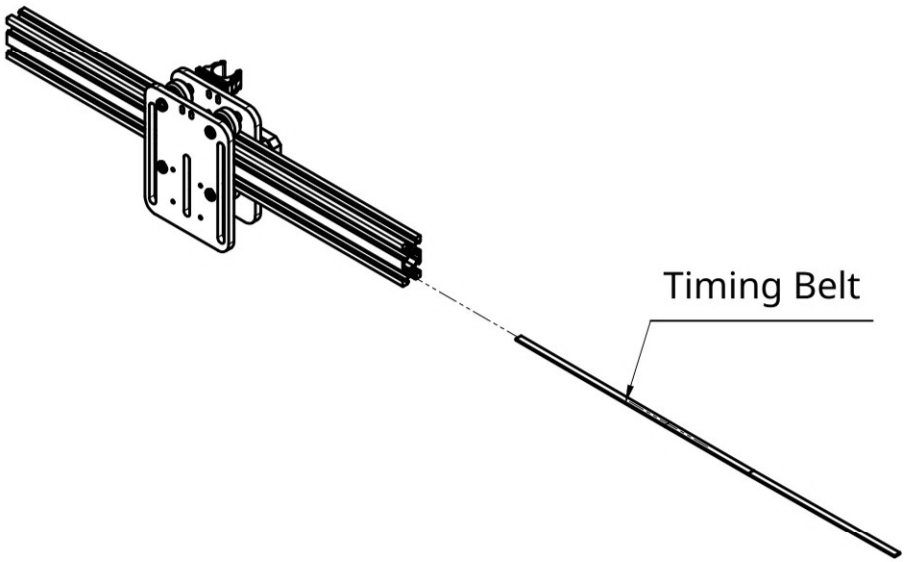
1


V-Slot Extrusion 2040

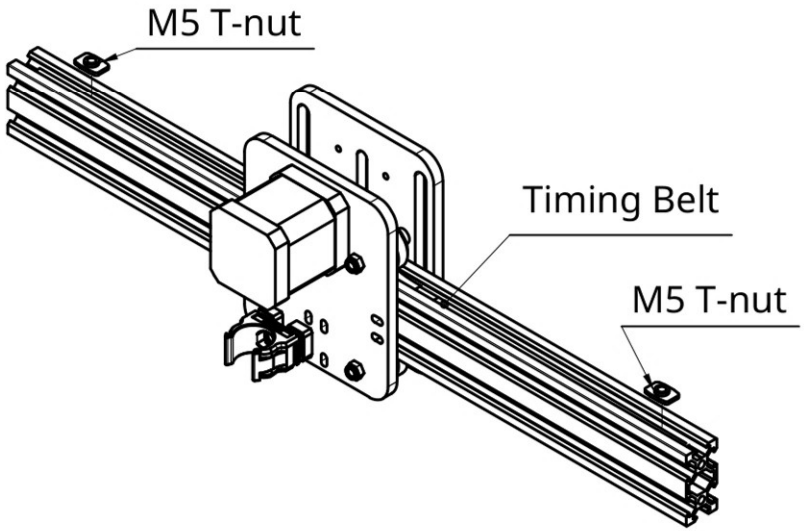
2


X-axis carriage

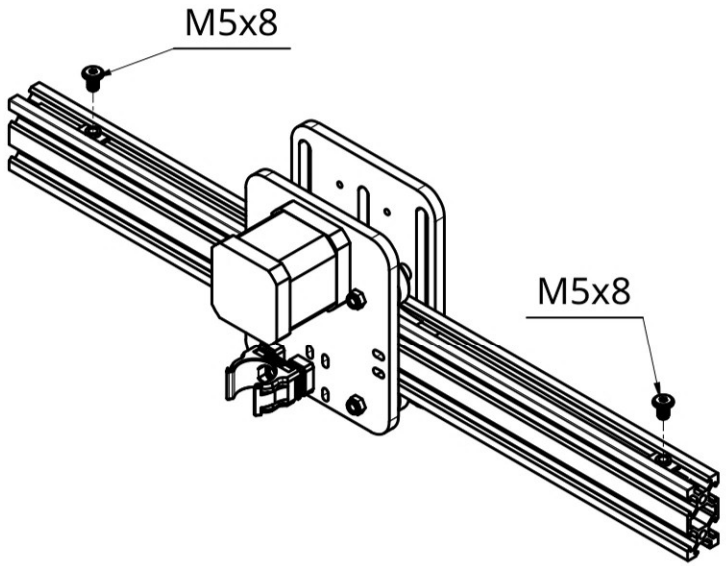
3


Timing Belt

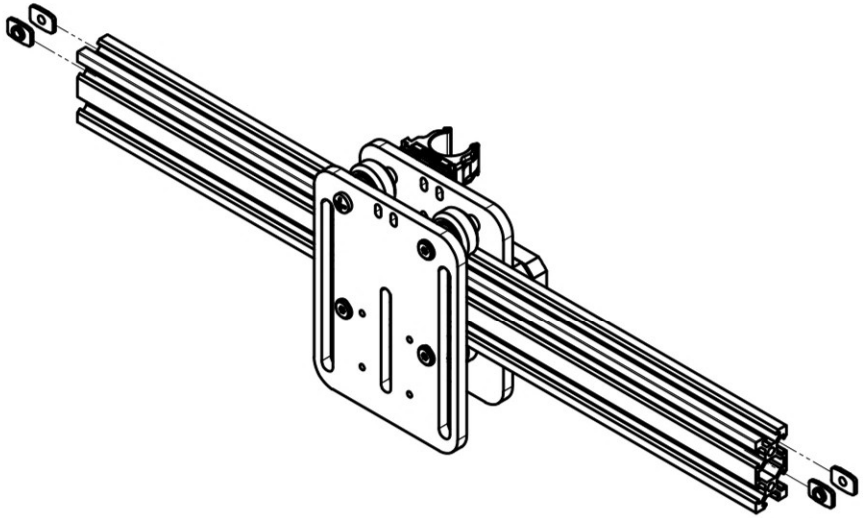
4


M5 T-nut
Timing Belt
M5 T-nut

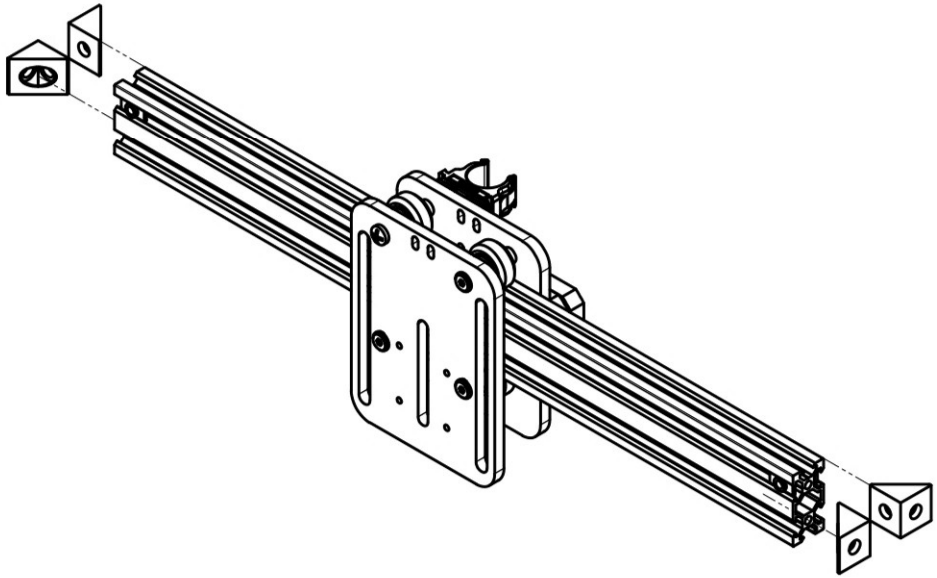
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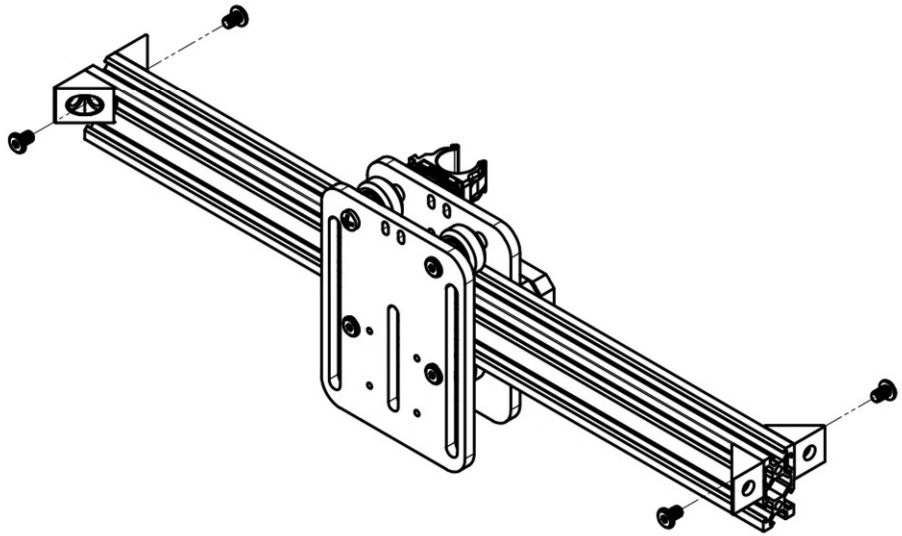
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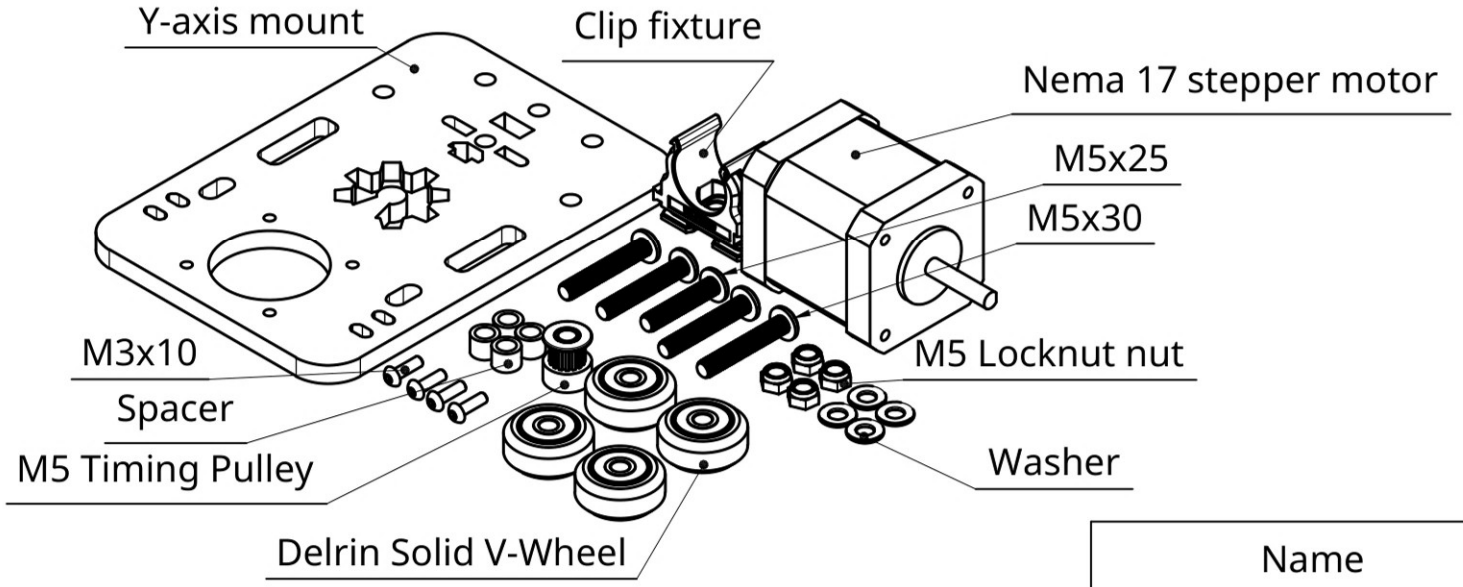
7



8



Y-Axis Assembly x2



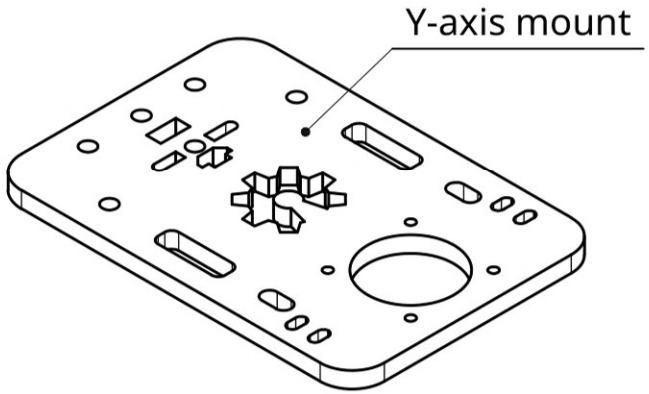
PARTS NEEDED (For One):

Name	Size	Quantity
Clip fixture	-	1
Washer	M5 Shim	4
Y-axis mount	-	1
Delrin Solid V-Wheel	-	4

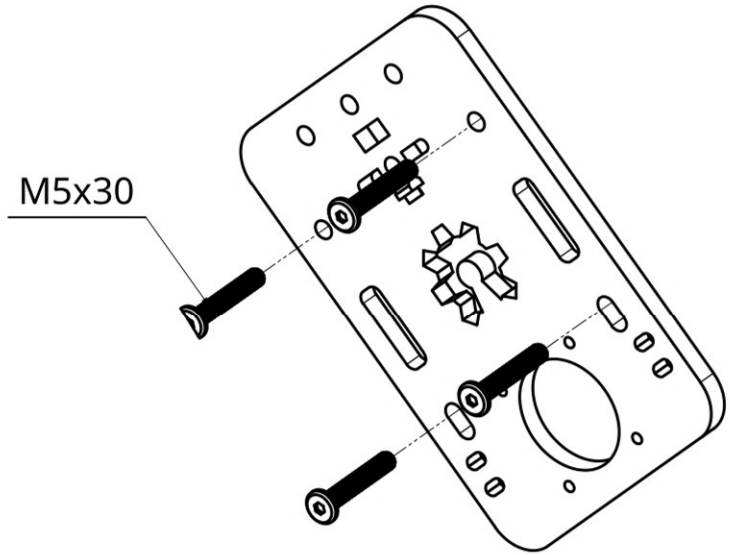
Name	Size	Quantity
Nylon Lock Nut	M5	3
30mm Ultra Low Profile Screw	M5	4
25mm Ultra Low Profile Screw	M5	1
Aluminium Spacer	5.1mm ID × 6mm	8
10mm Ultra Low Profile Screw	M3	4
Nema 17 stepper motor	42x42mm	1
GT2 Pulley 20T	M5	1

Y-Axis Cariage

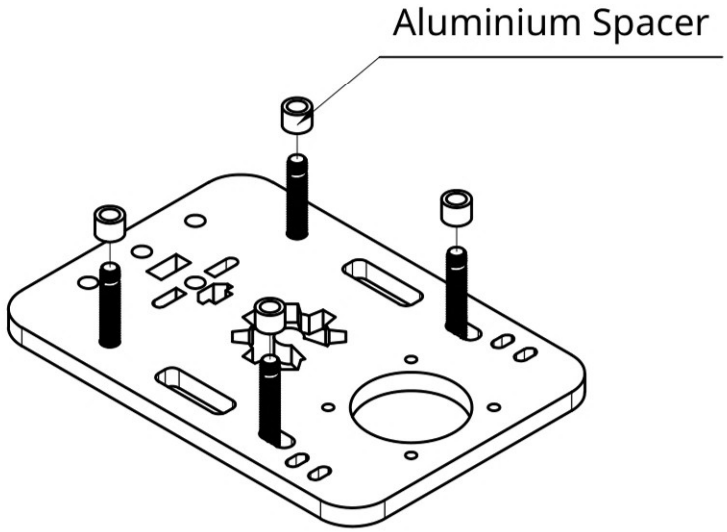
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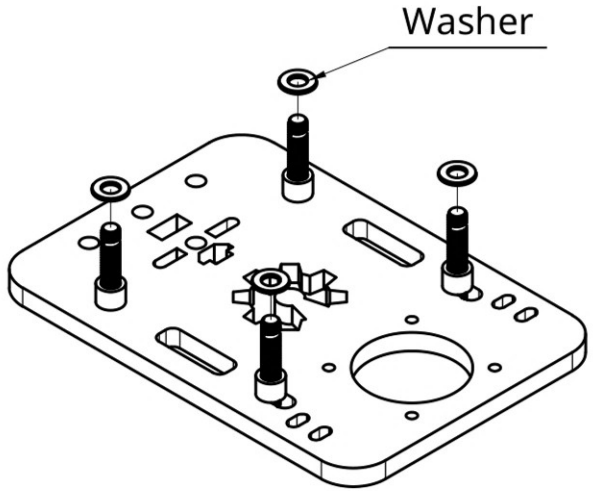
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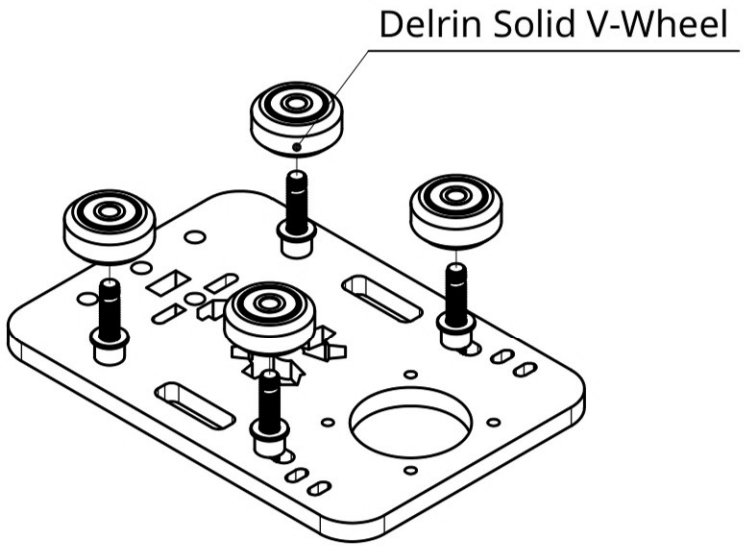
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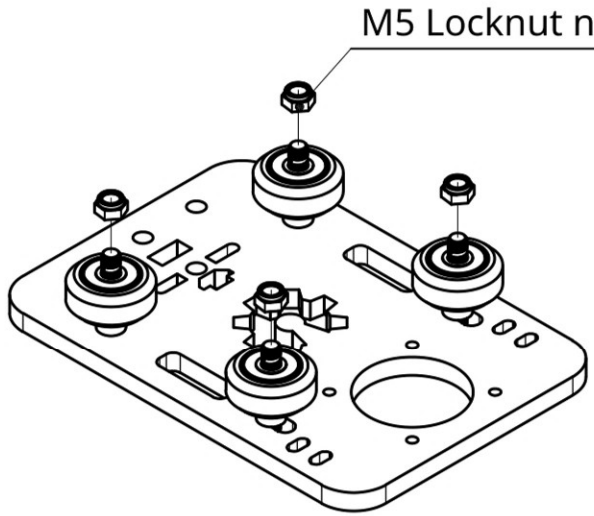
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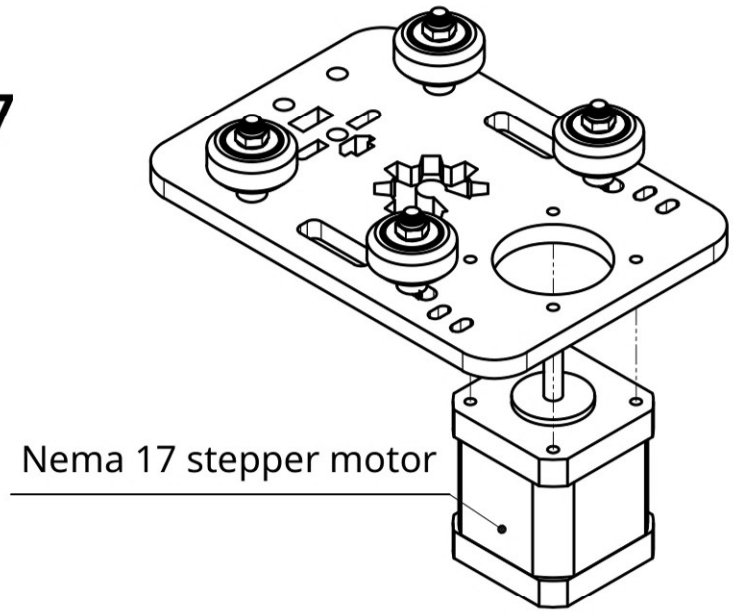
5


Delrin Solid V-Wheel

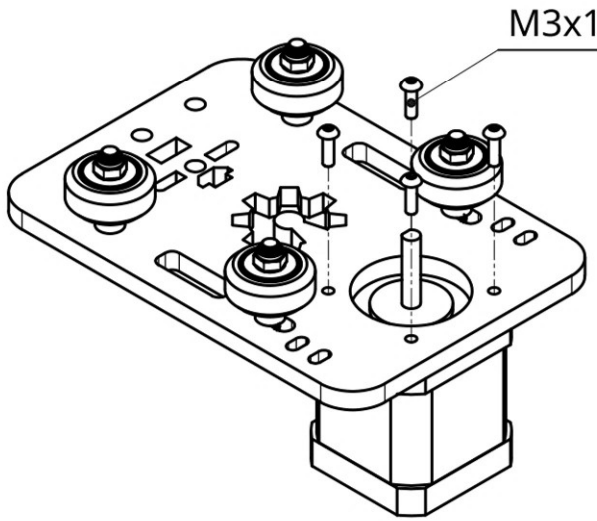
6


M5 Locknut nut

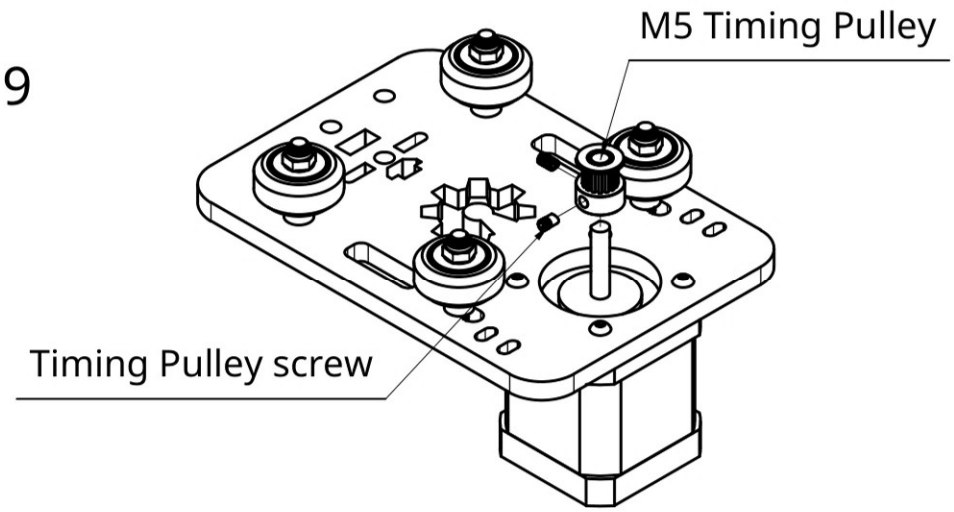
7


Nema 17 stepper motor

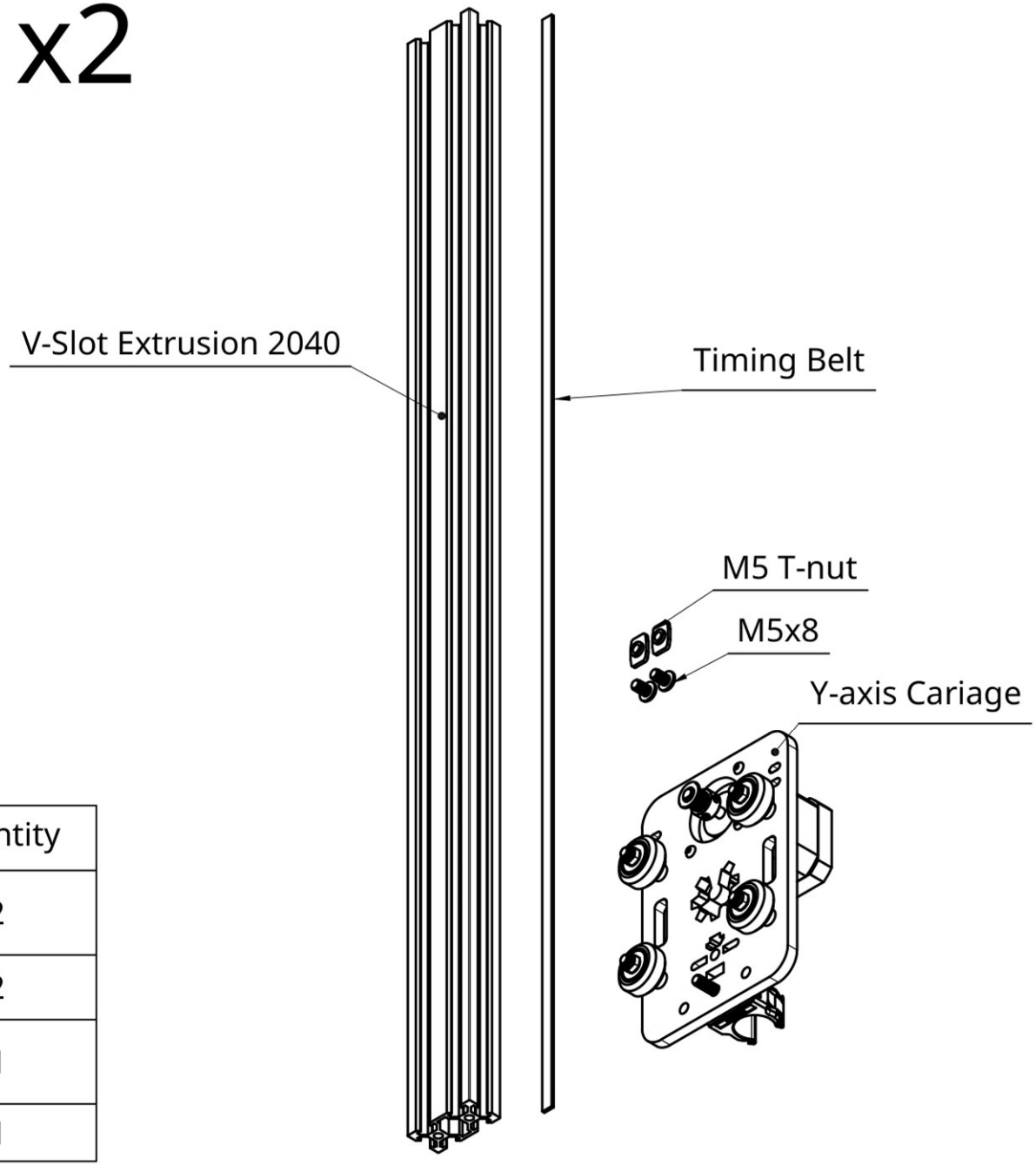
8


M3x10

9



Y-Axis Assembly x2

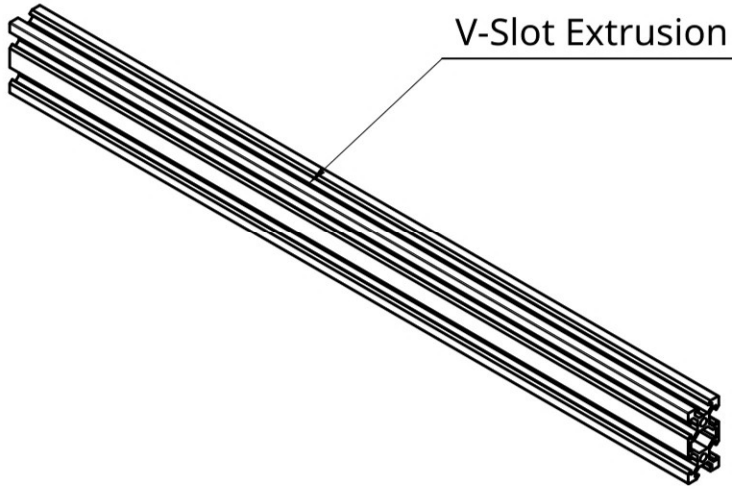


PARTS NEEDED :

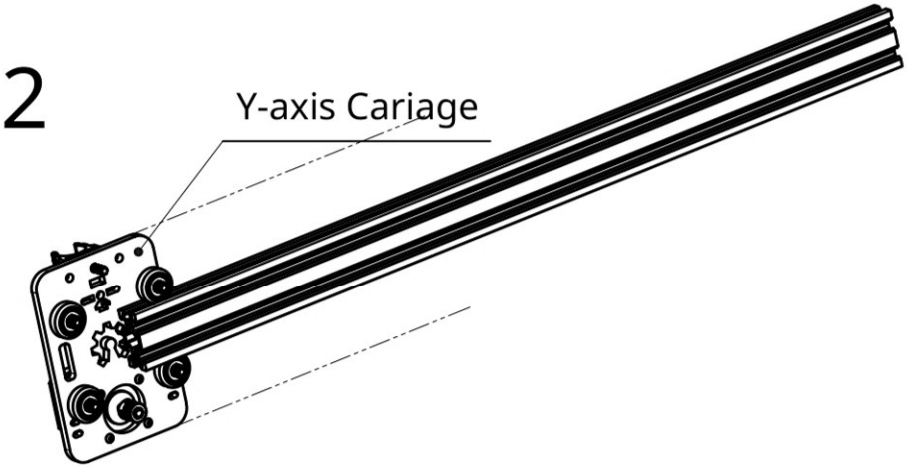
Name	Size	Quantity
8mm Ultra Low Profile Screw	M5	2
Drop-In T-Nut	M5	2
V-Slot Extrusion 2040	610 mm	1
GT2 Timing Belt	650 mm	1

Y-Axis Assembly

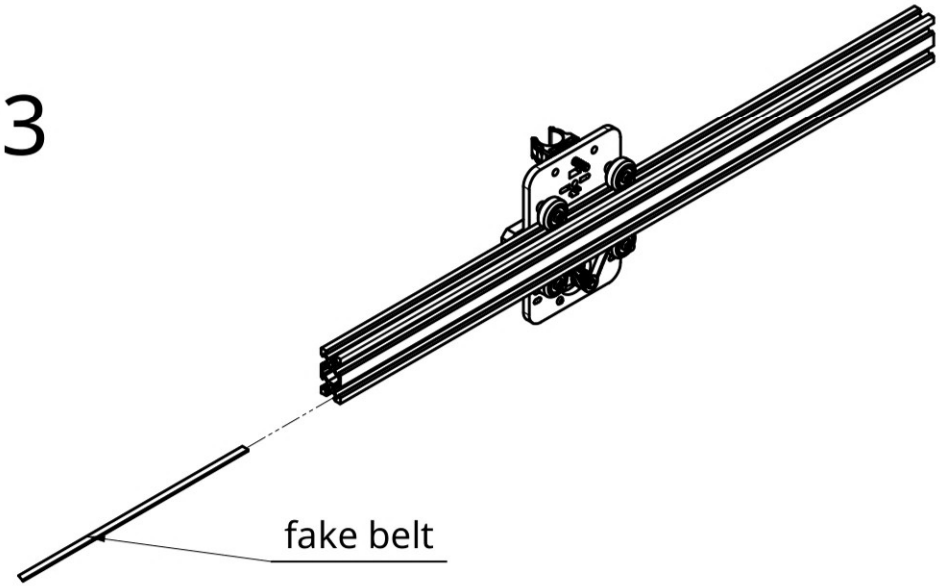
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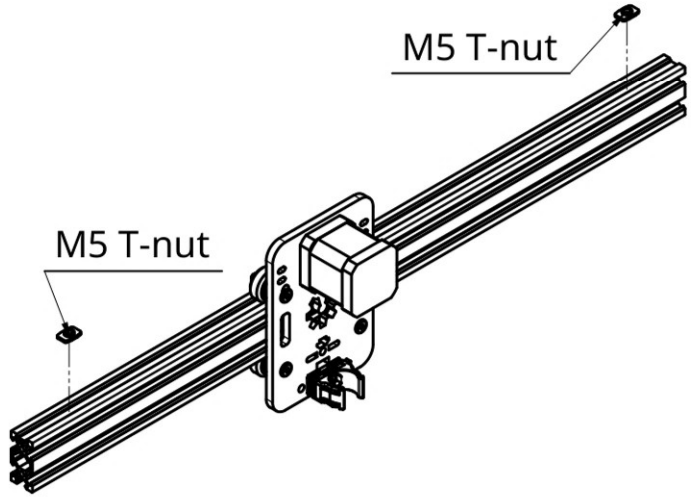
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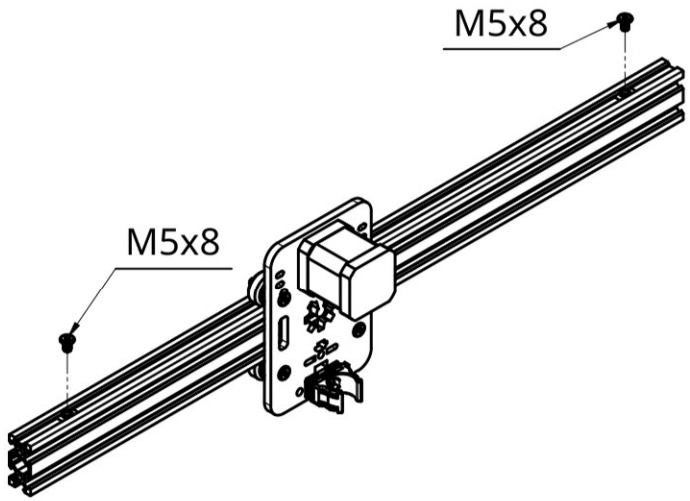
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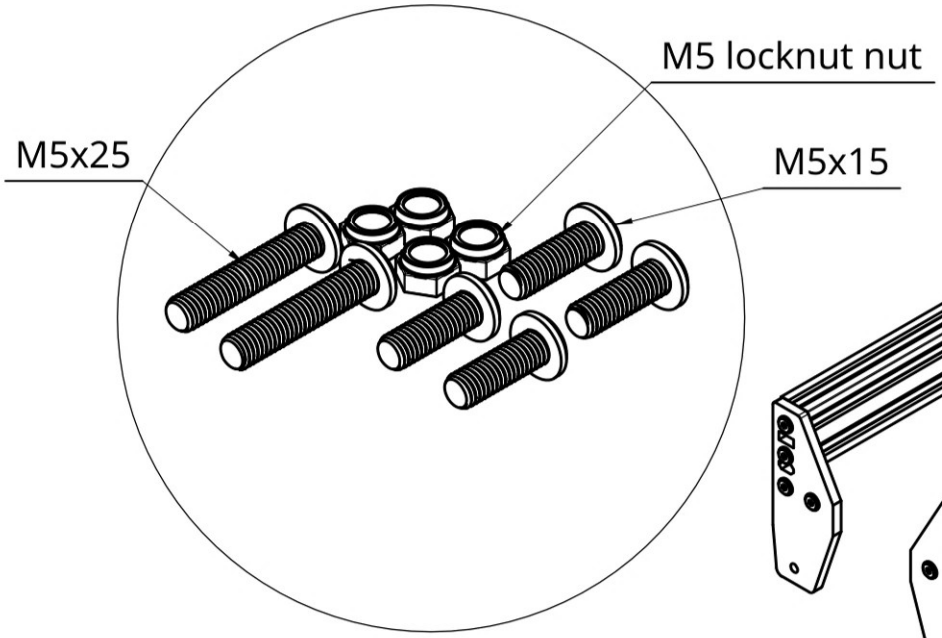
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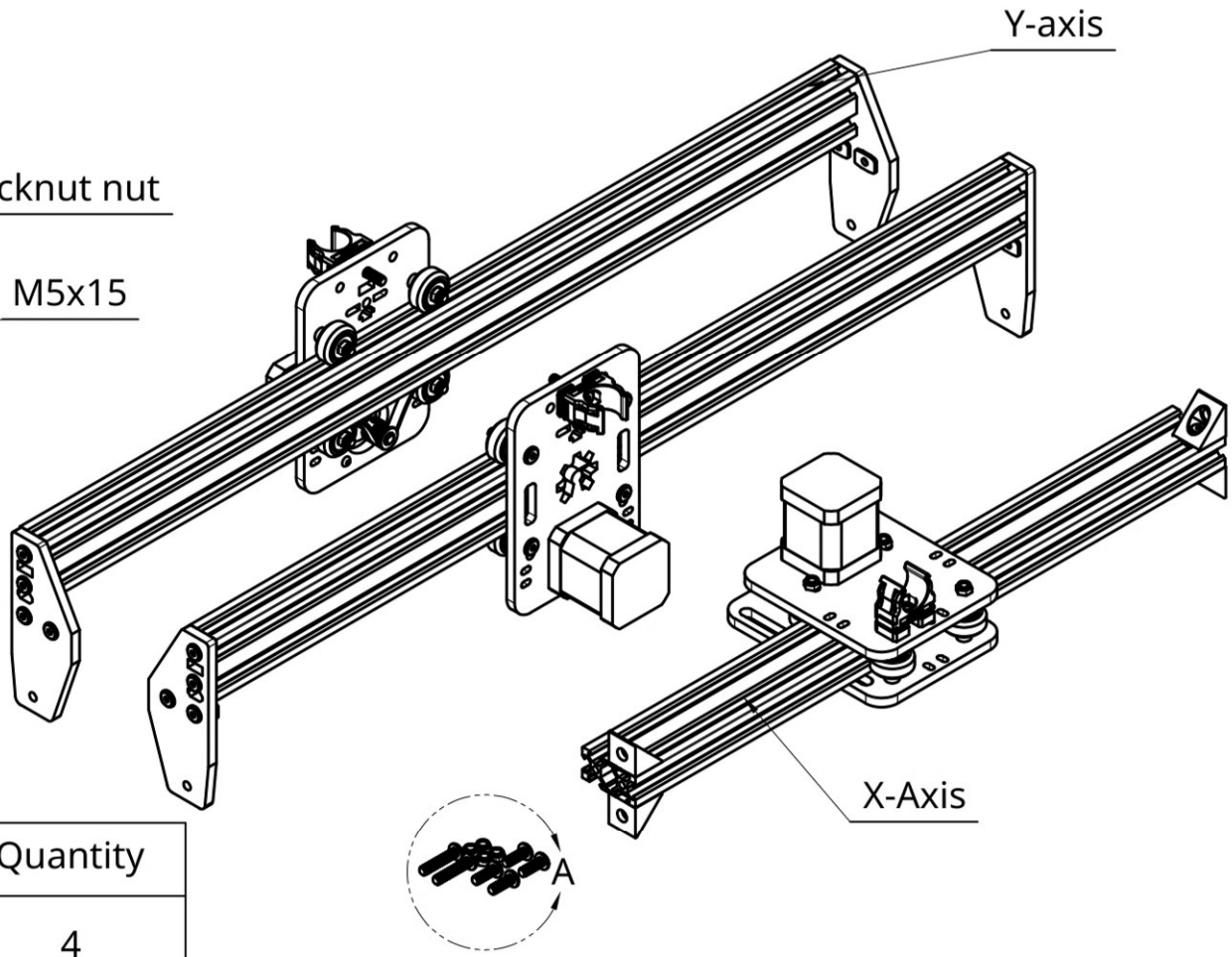


5



Axis Assembly



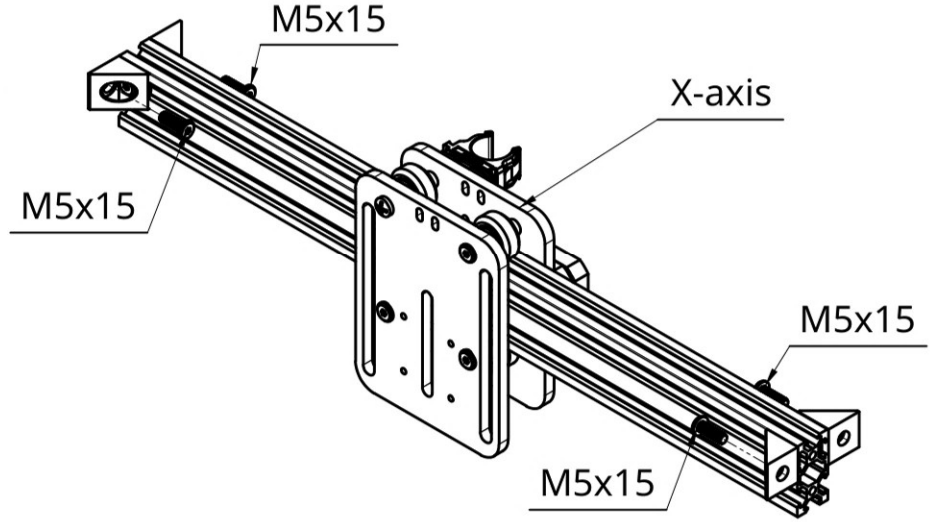


PARTS NEEDED :

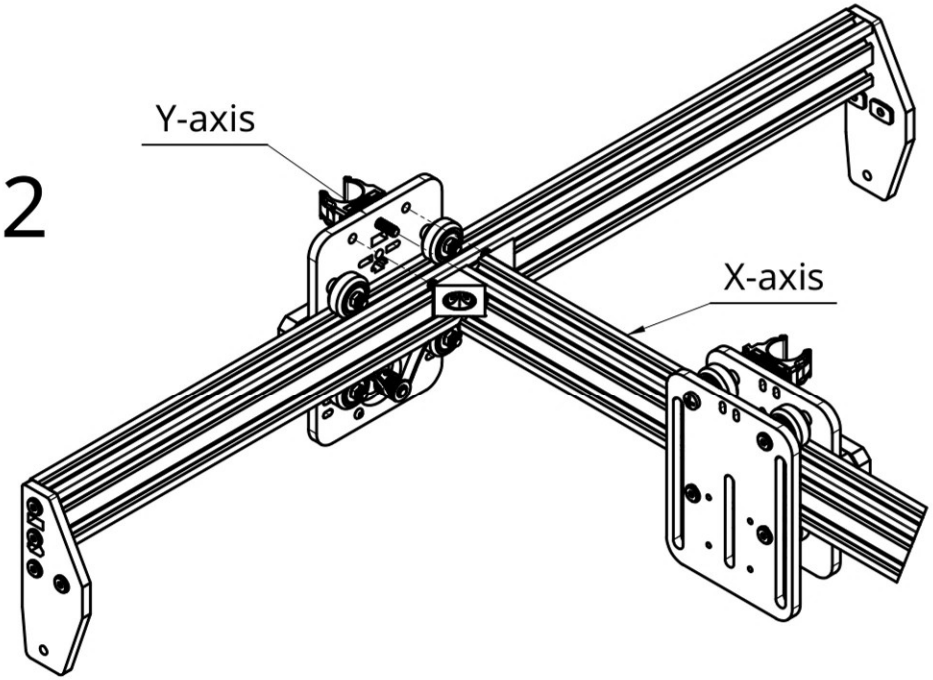
Name	Size	Quantity
15mm Ultra Low Profile Screw	M5	4
25mm Ultra Low Profile Screw	M5	2
locknut nut	M5	4
Y-axis Assembly	-	2
X-Aixs Assembly	-	1

Axis Assembly

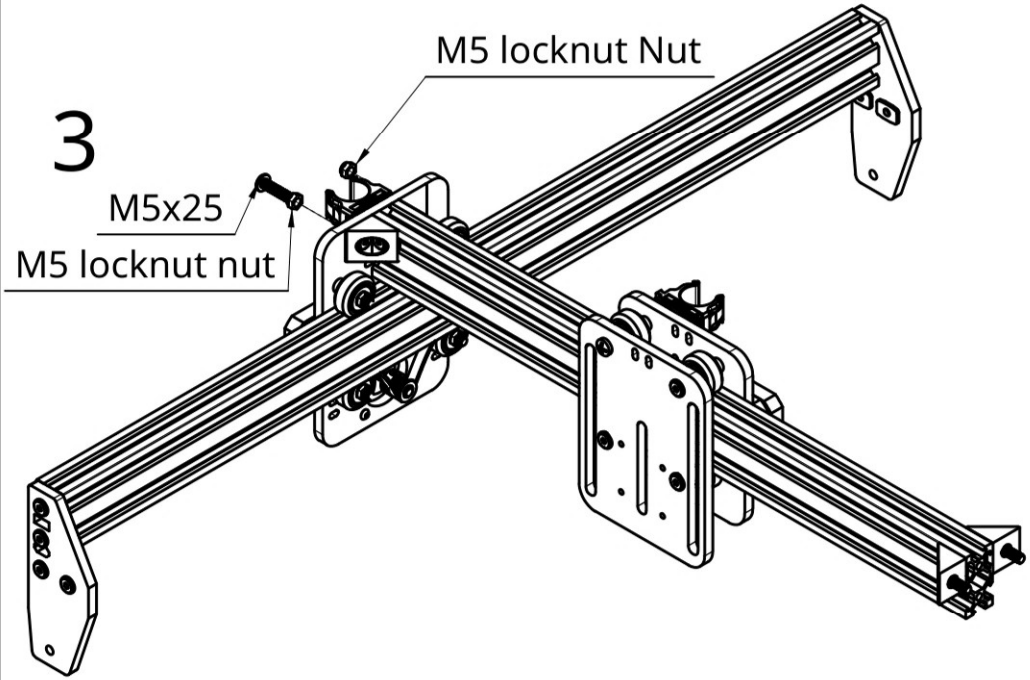
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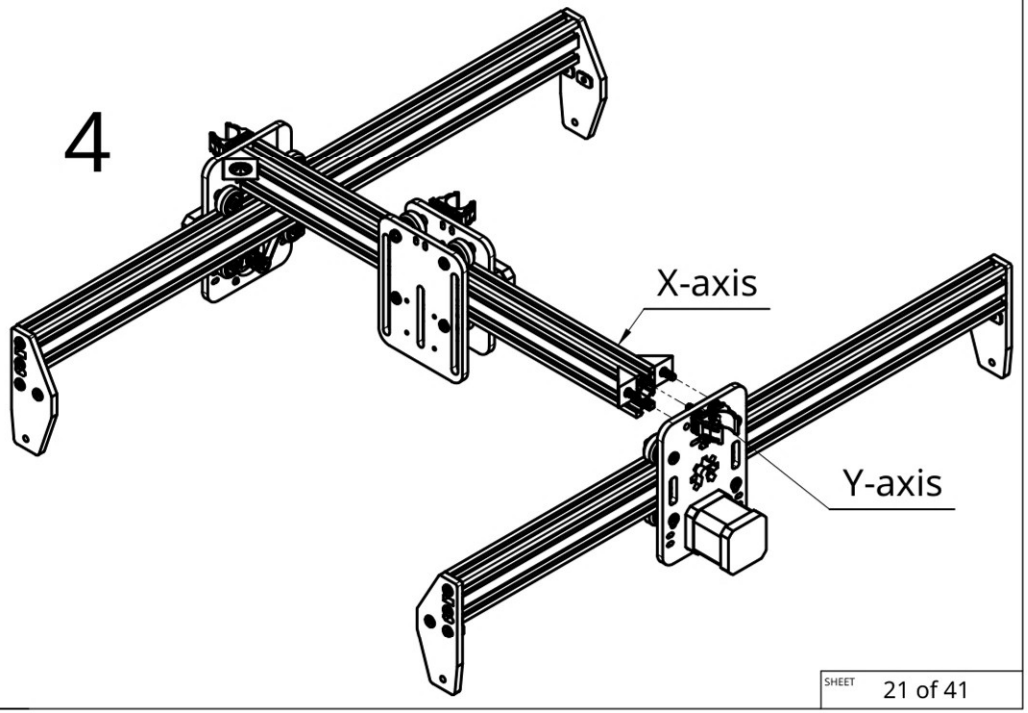
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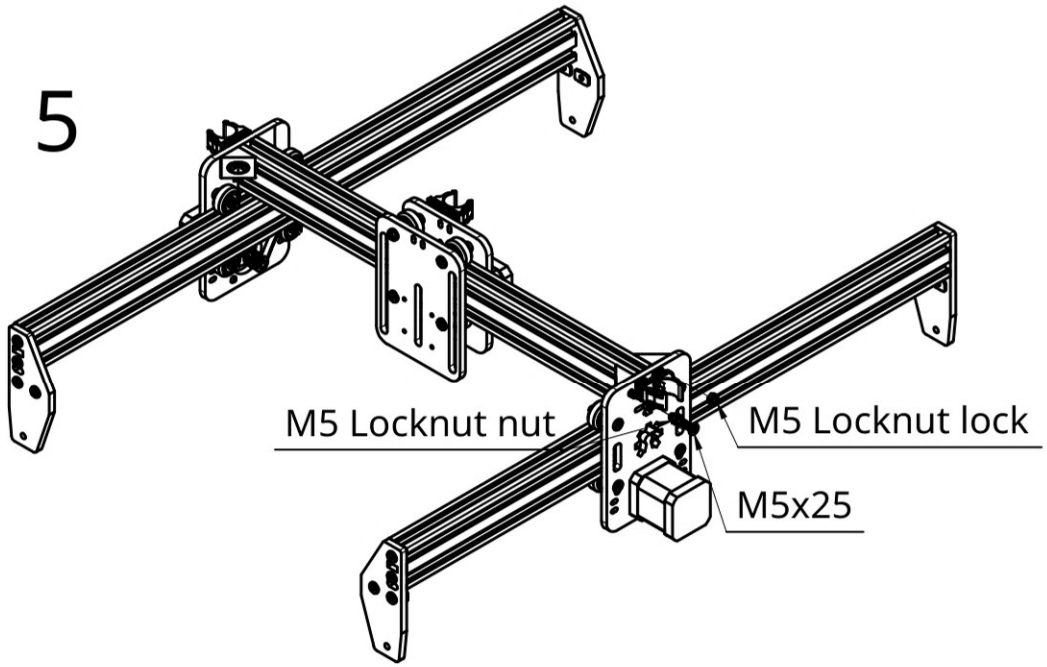


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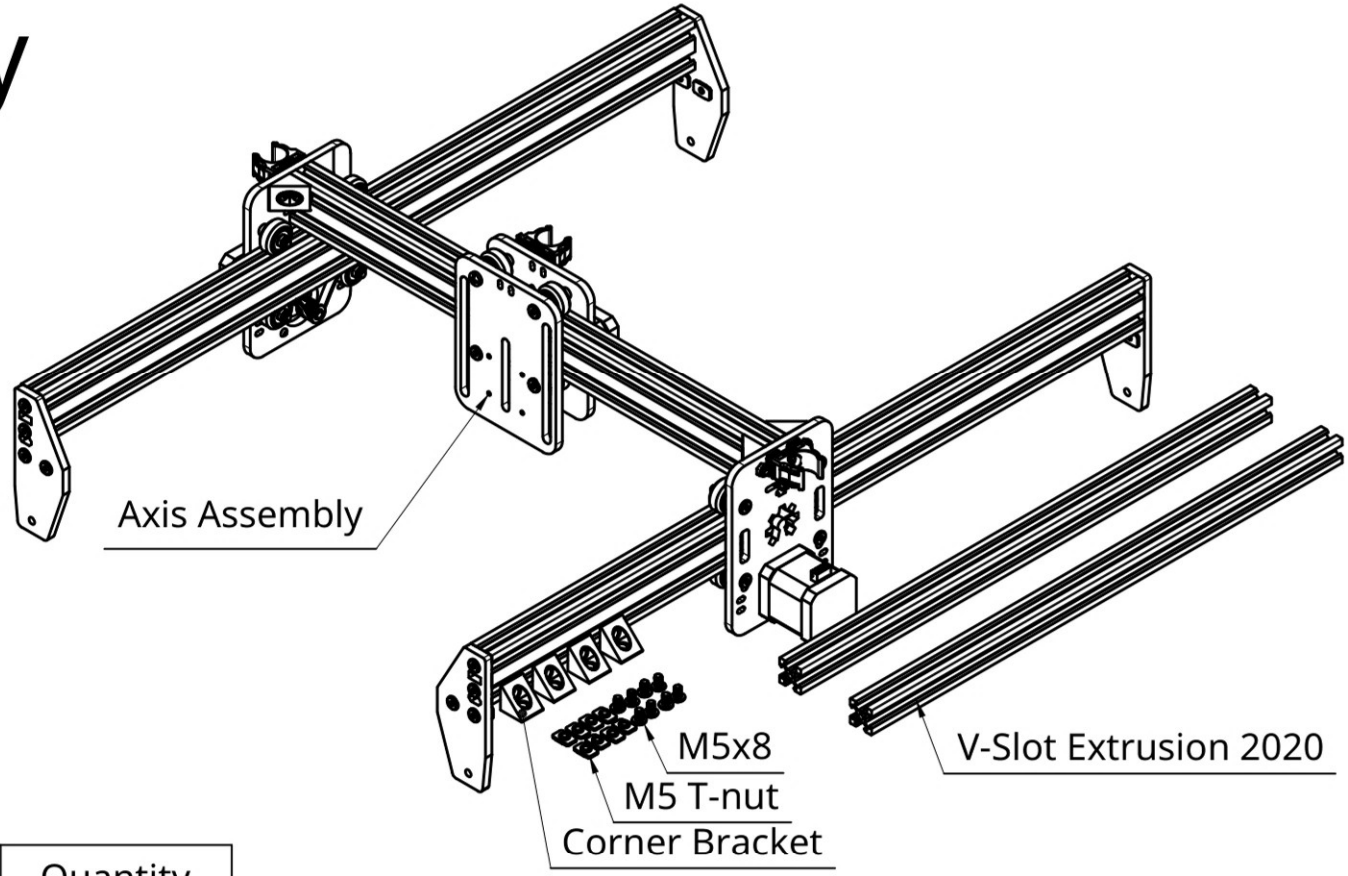


1:6

5



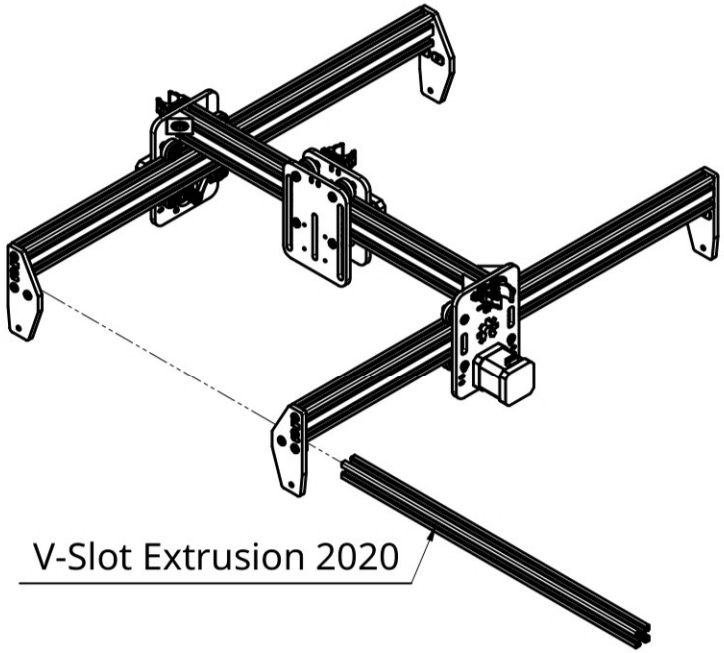
Axis Assembly



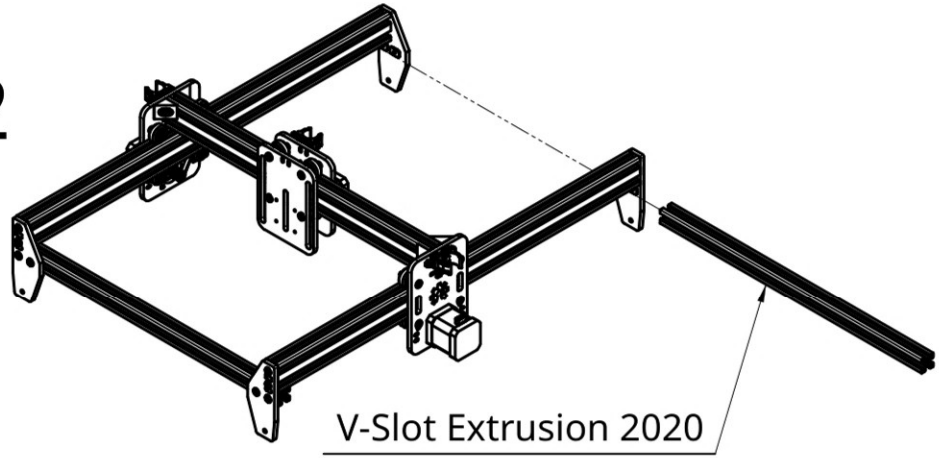
PARTS NEEDED :

Name	Size	Quantity
8mm Ultra Low Profile Screw	M5	8
Drop-In T-Nut	M5	8
Corner Bracket	20 Series	4
Axis Assembly	-	1
V-Slot Extrusion 2020	440 mm	2

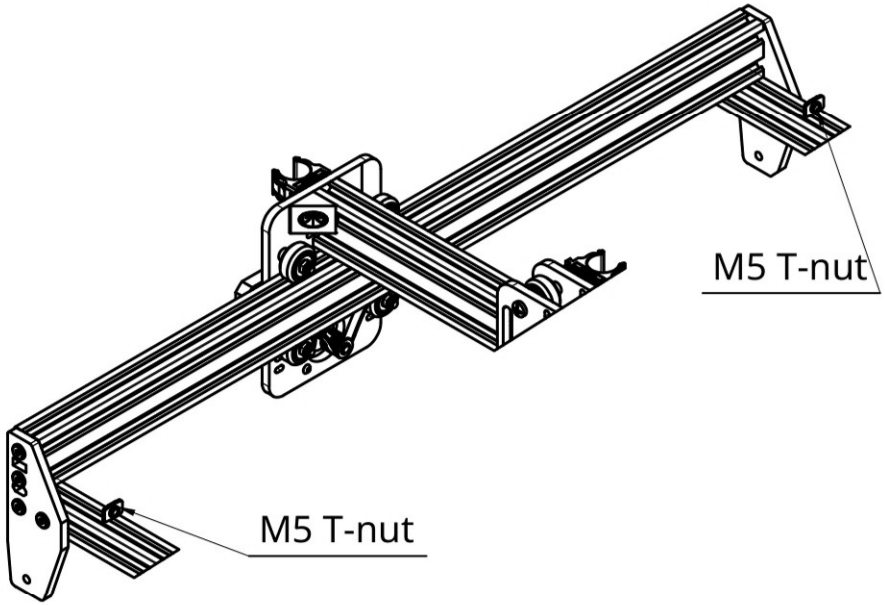
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V-Slot Extrusion 2020

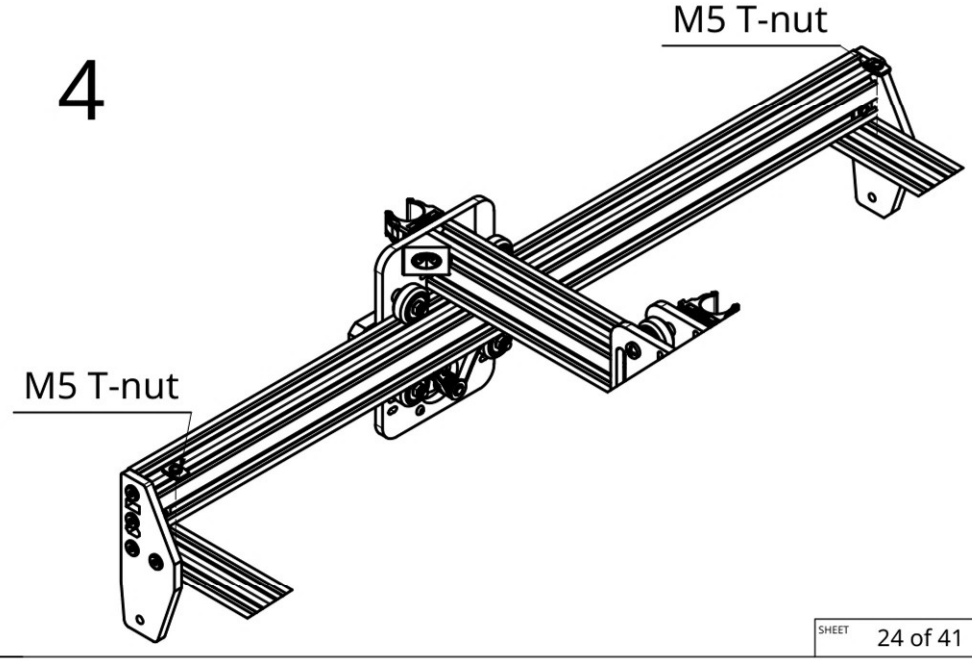
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V-Slot Extrusion 2020

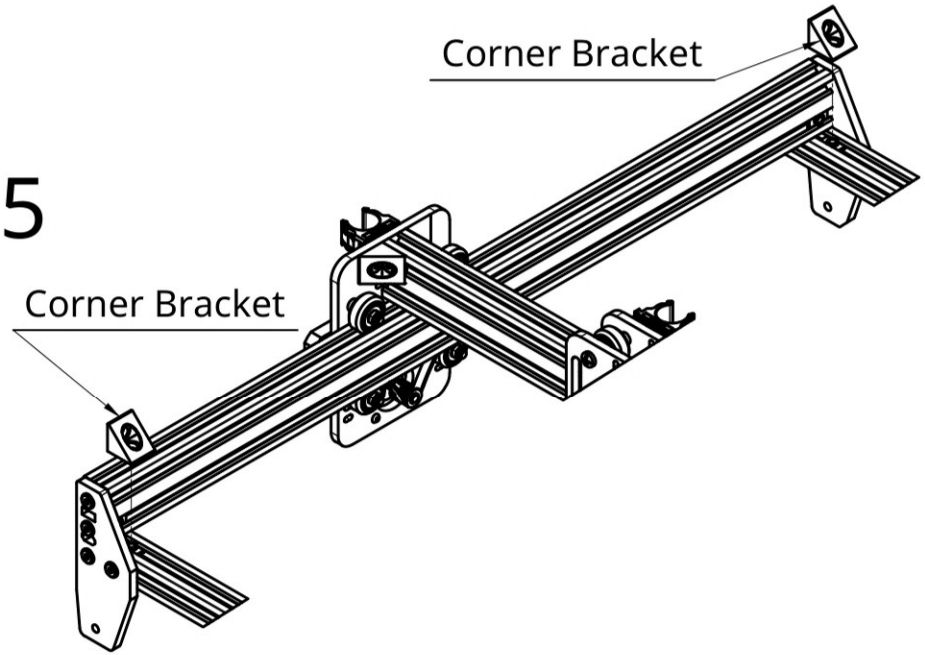
3


M5 T-nut
M5 T-nut

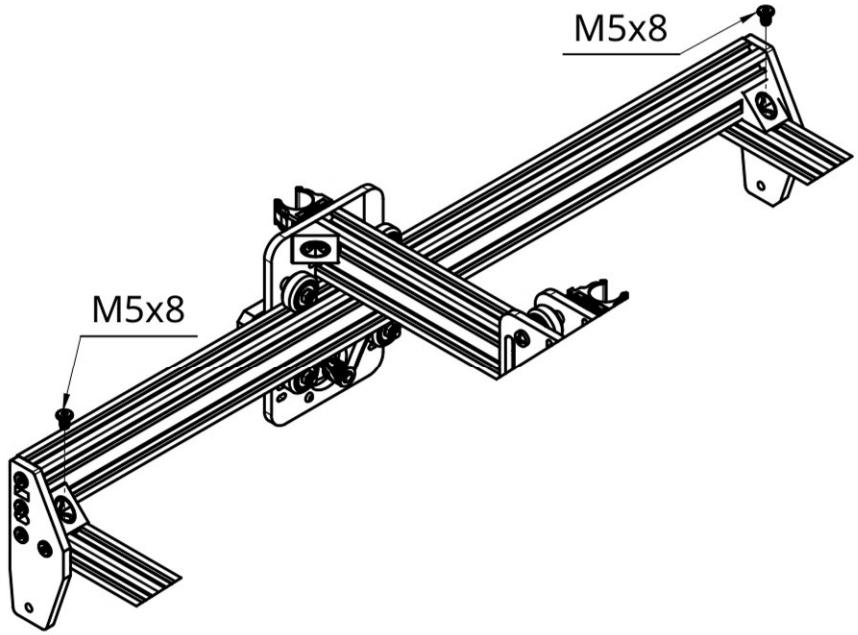
4


M5 T-nut
M5 T-nut

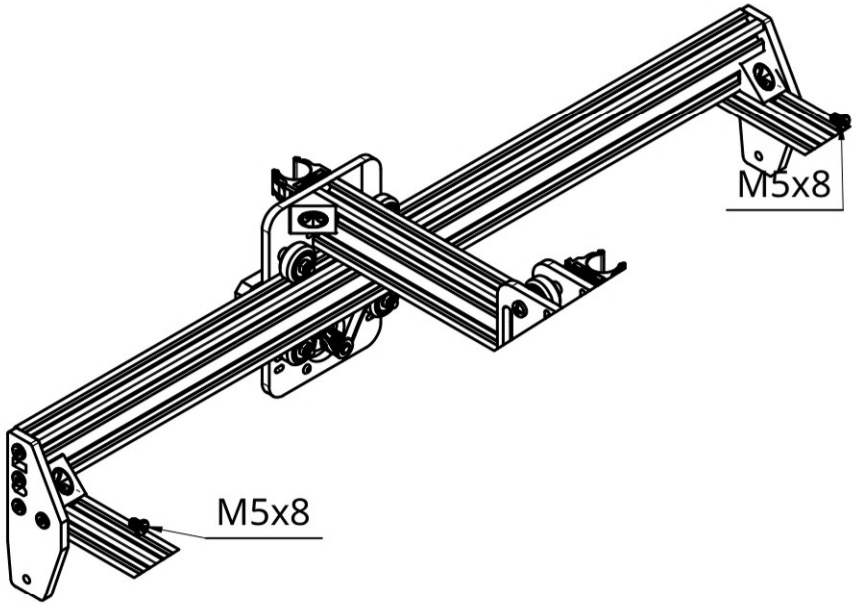
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Corner Bracket
Corner Bracket

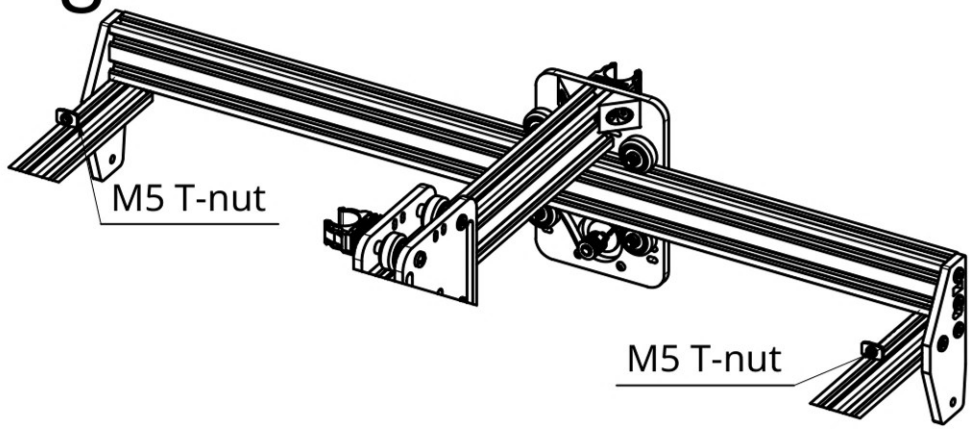
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M5x8
M5x8

7

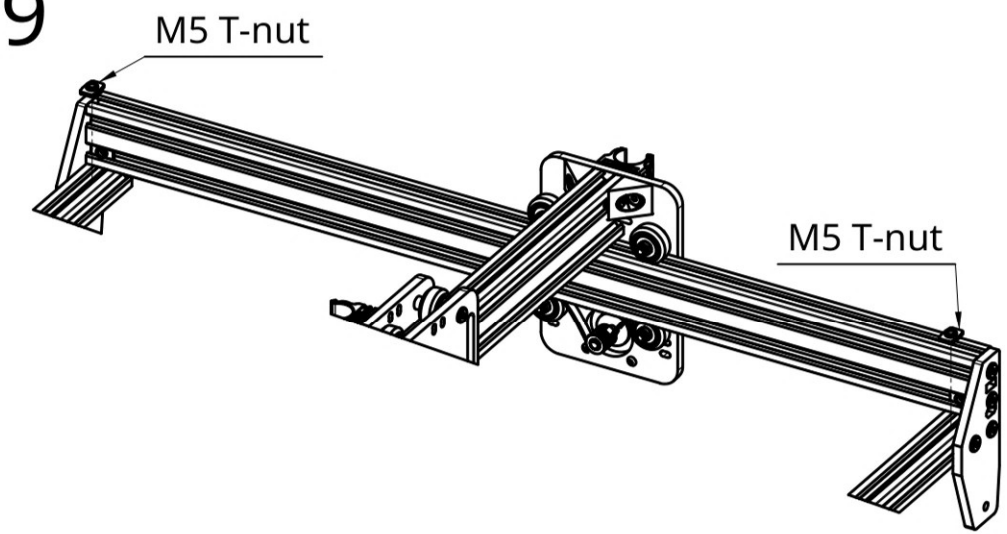

M5x8
M5x8

8

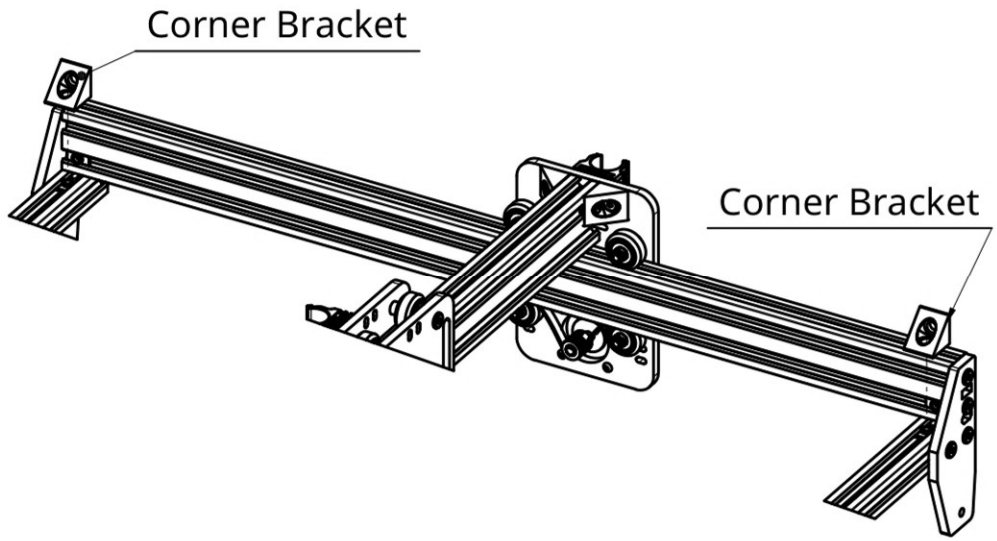

M5 T-nut
M5 T-nut

1:4.8

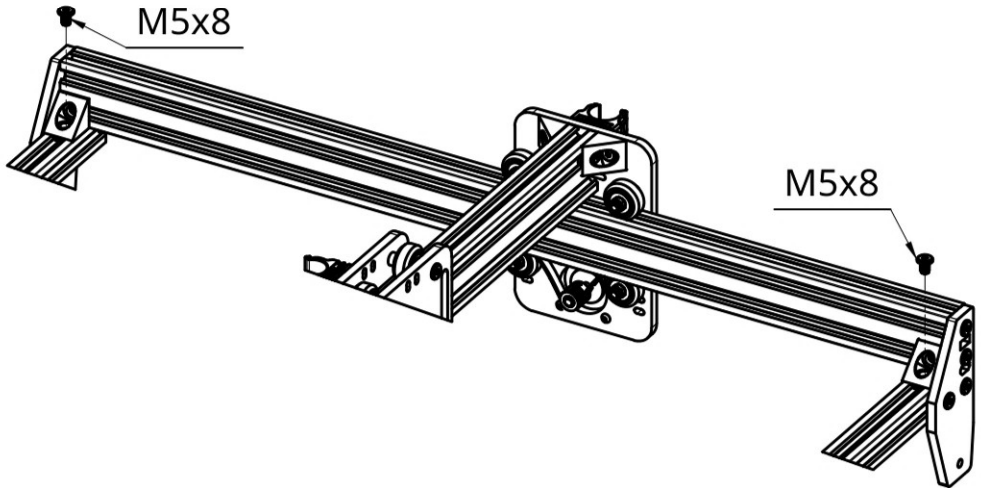
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M5 T-nut
M5 T-nut

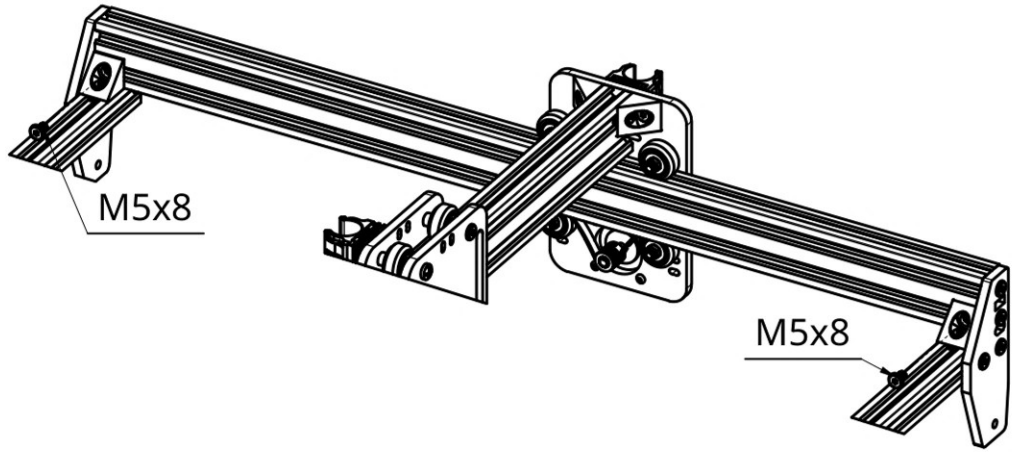
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Corner Bracket
Corner Bracket

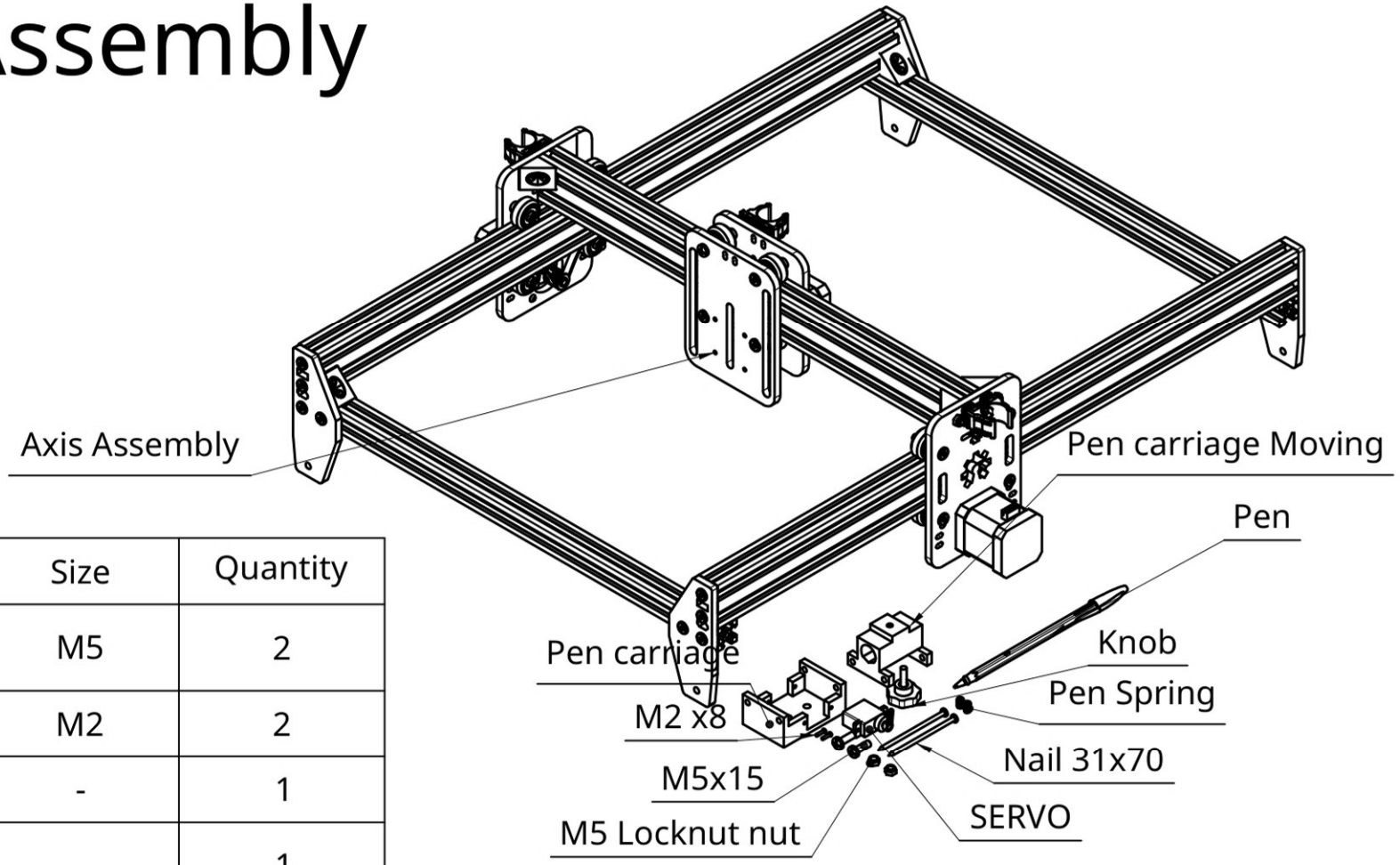
11


M5x8
M5x8

12


M5x8
M5x8

Pen Lift Assembly

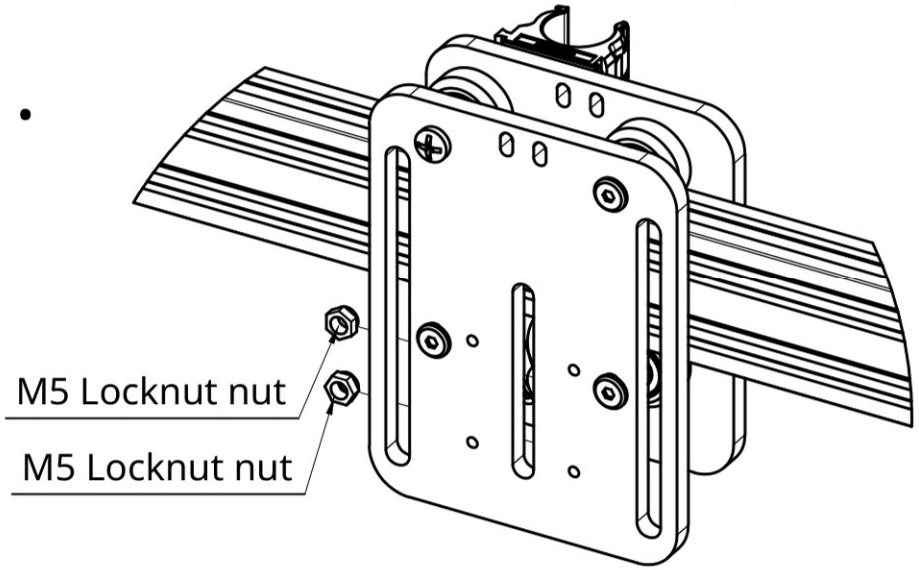


PARTS NEEDED :

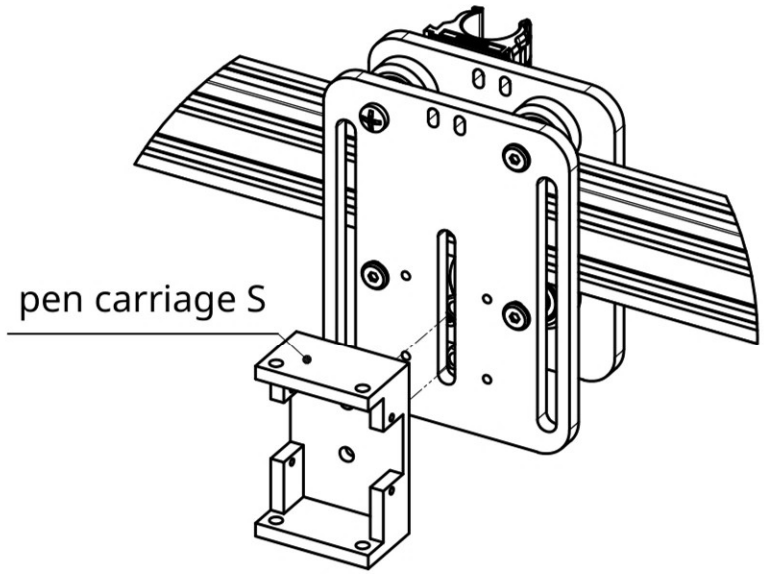
Name	Size	Quantity
15mm Ultra Low Profile Screw	M5	2
8mm Screw	M2	2
Pen carriage	-	1
Pen carriage Moving	-	1
locknut nut	M5	2
9g Servo Motor 180°	-	1
Concrete Nail	70mm	2
Screw Knobs	M5	1
Pen Springs	7.5mm	2

Pen Lift Assembly

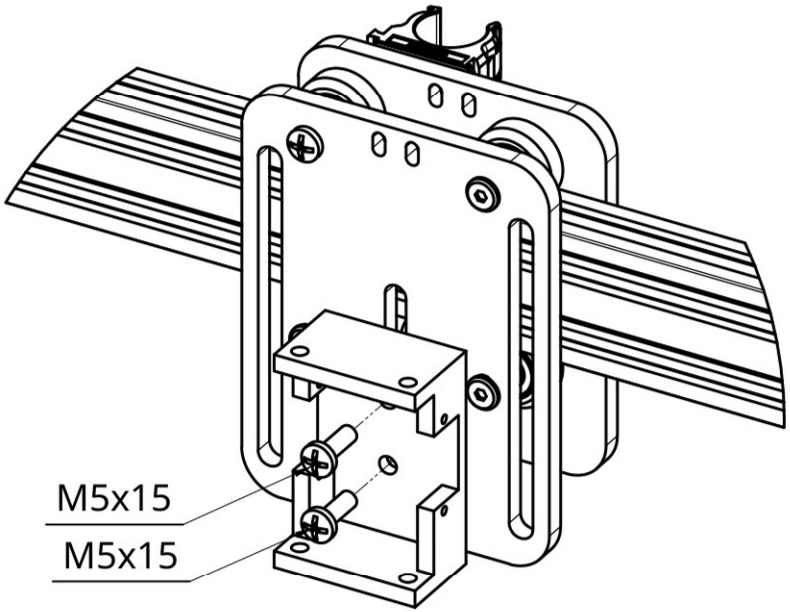
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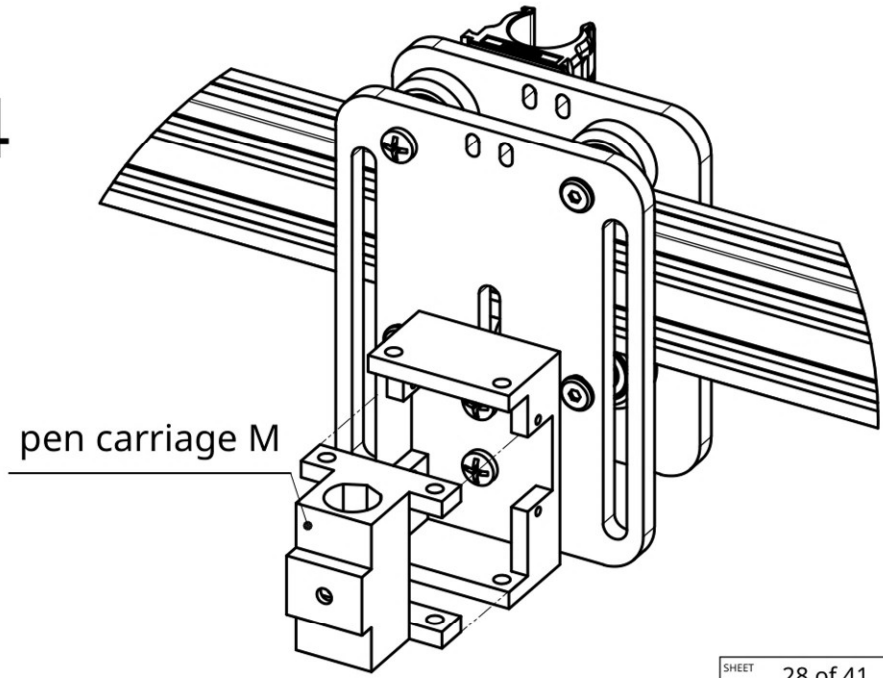
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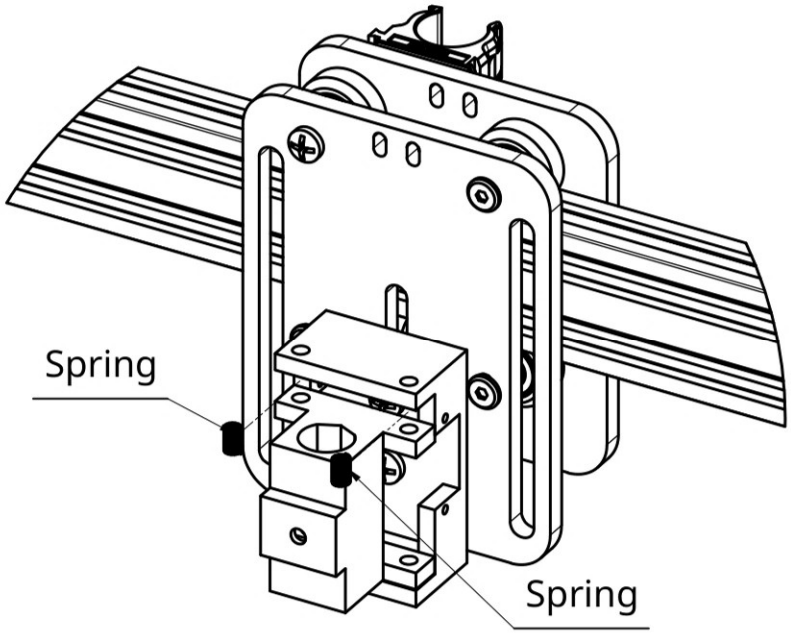
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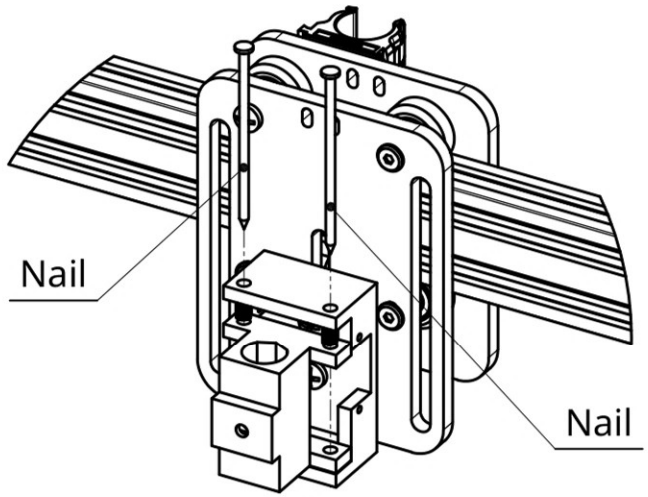
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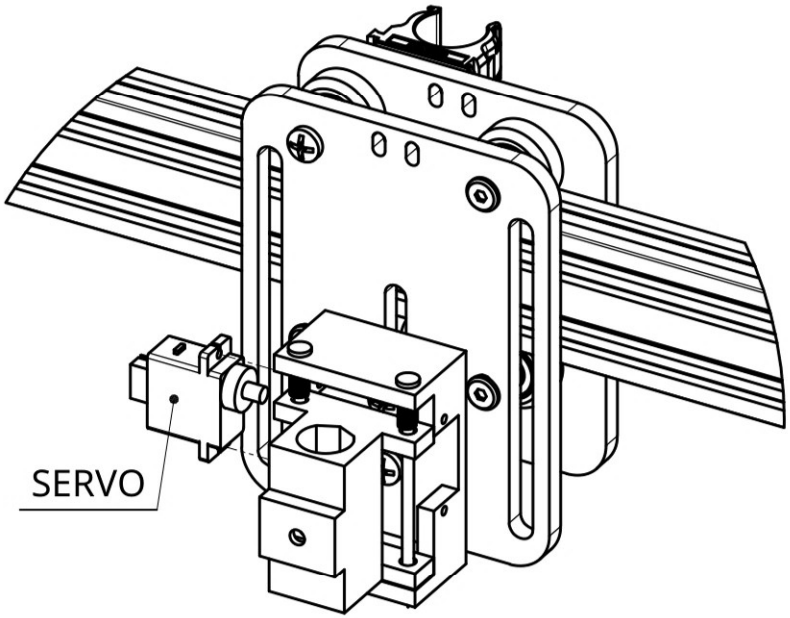
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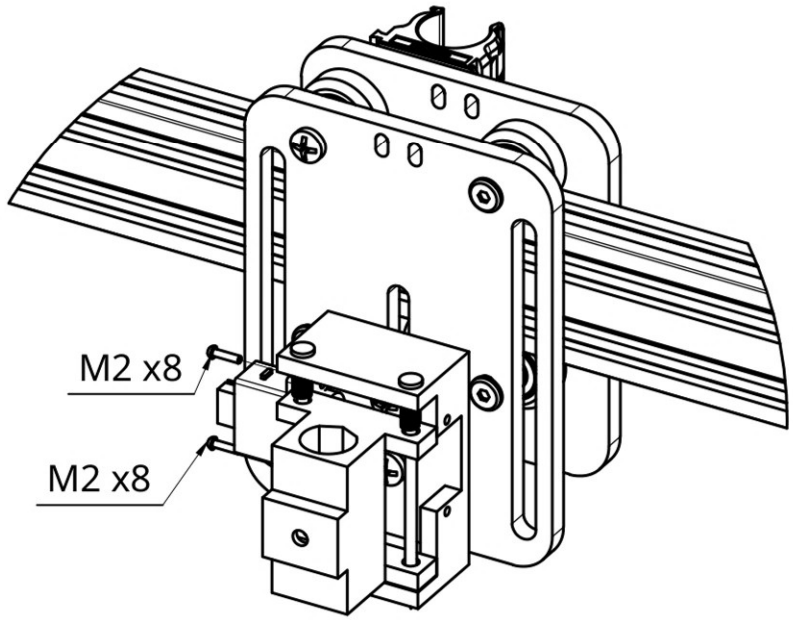
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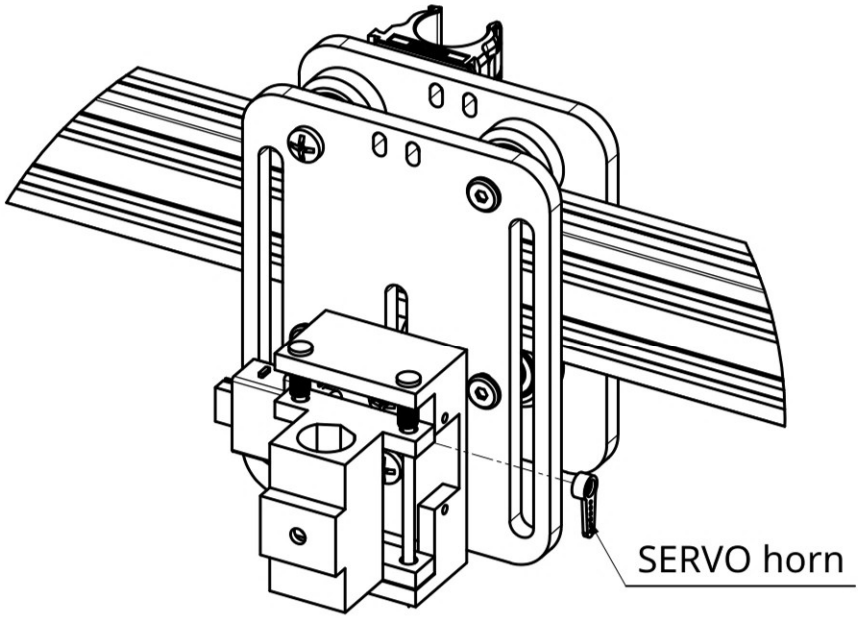
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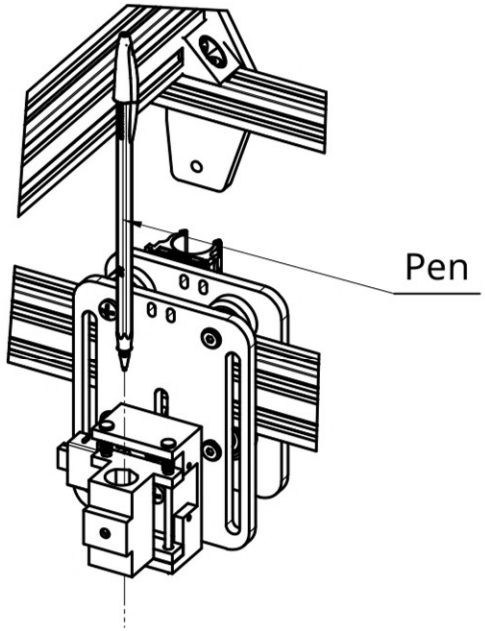
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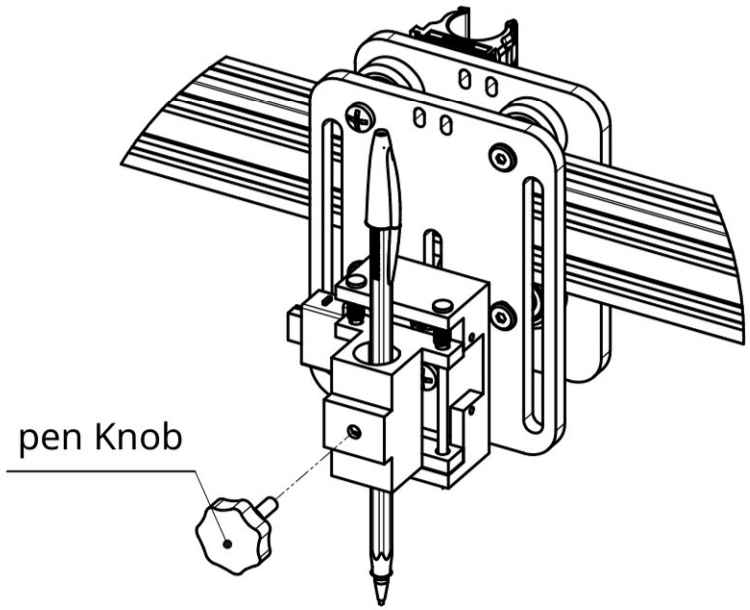
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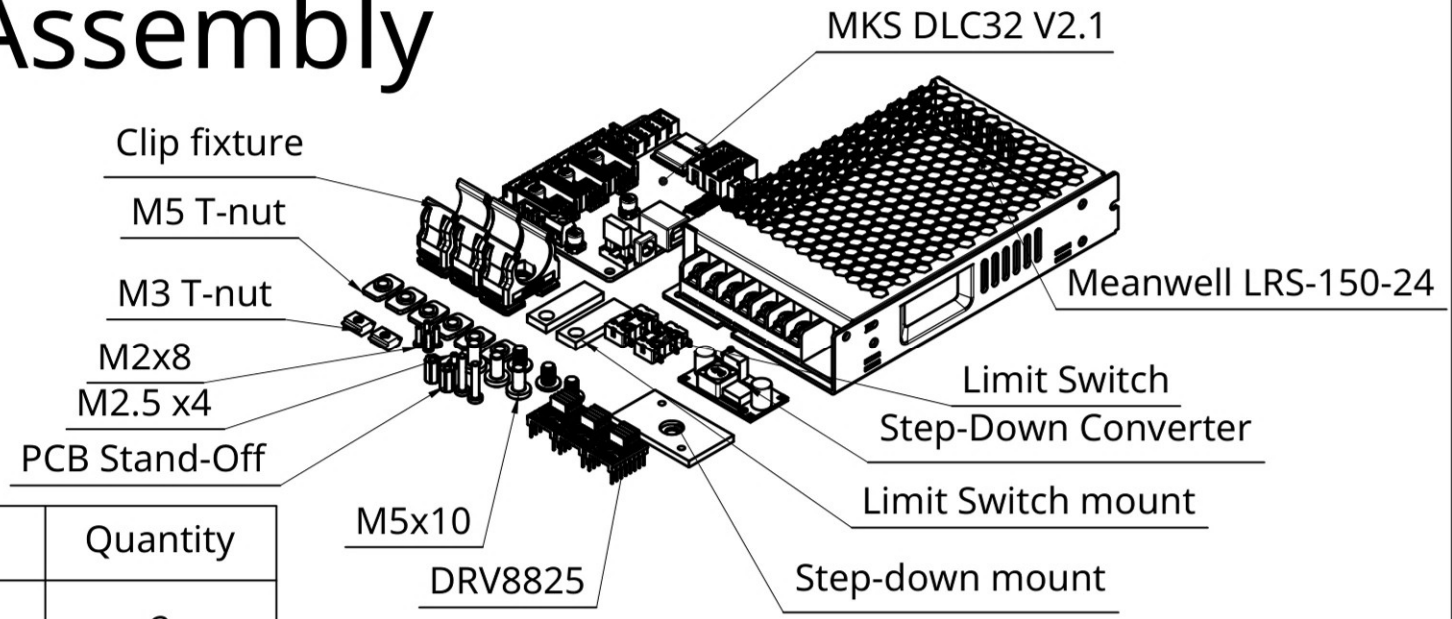
10



11



Electronics Assembly



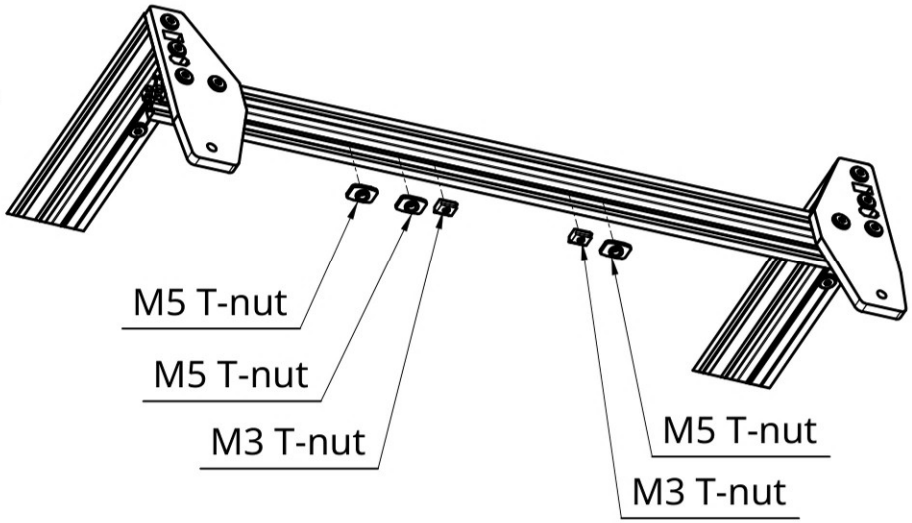
PARTS NEEDED :

Name	Size	Quantity
8mm Ultra Low Profile Screw	M5	2
10mm Ultra Low Profile Screw	M5	3
8mm Screw	M2	4
PCB Stand-Off	M3	2
Drop-In T-Nut	M5	5
Drop-In T-Nut	M3	2
Meanwell LRS-150-24	-	1
16mm Screw	M3	2
MKS DLC32 V2.1	-	1
Clip fixture	-	3

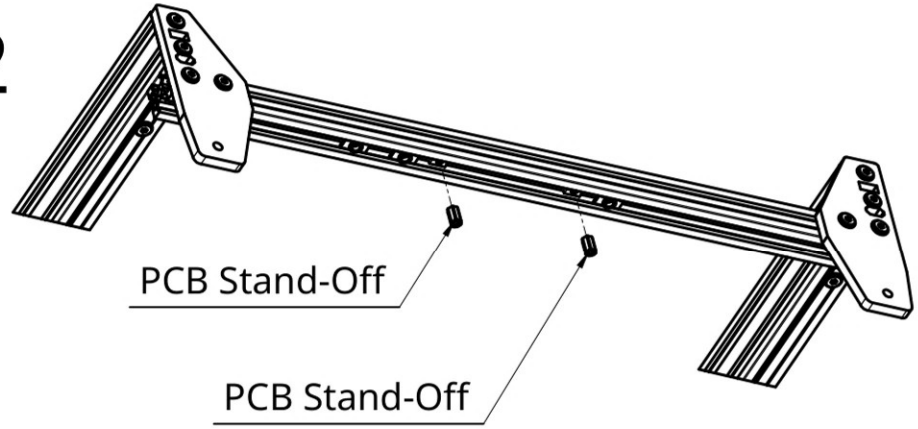
Limit Switch	-	2
Limit Switch mount	-	2
DRV8825 Motor Driver	-	3
XL6009 DC-DC Boost-Buck Converter Module	-	1
Boost-Buck Converter Module mount	-	1
4mm Screws	M2.5	2

Electronics Assembly

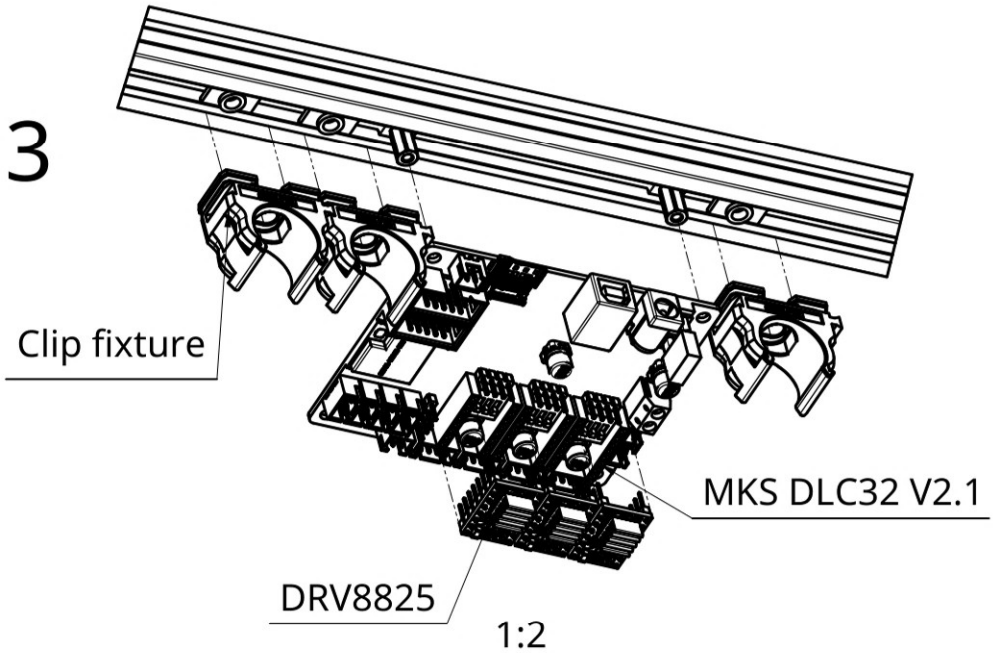
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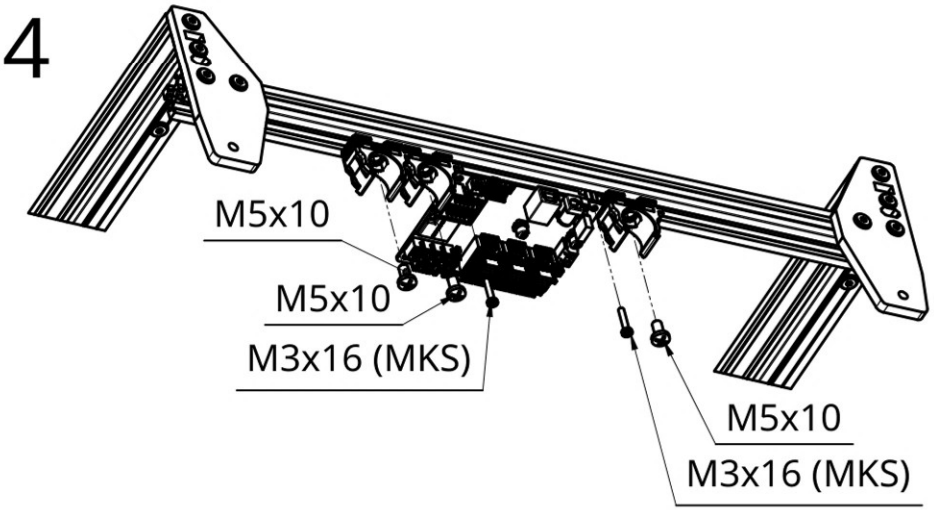


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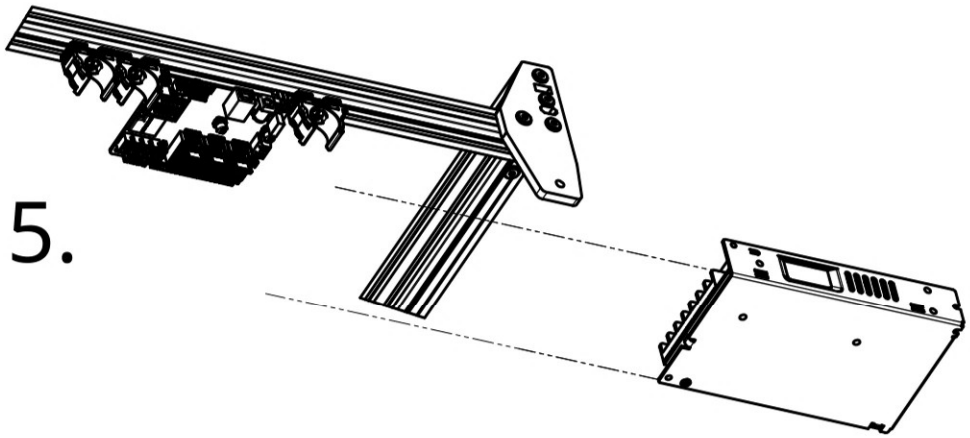


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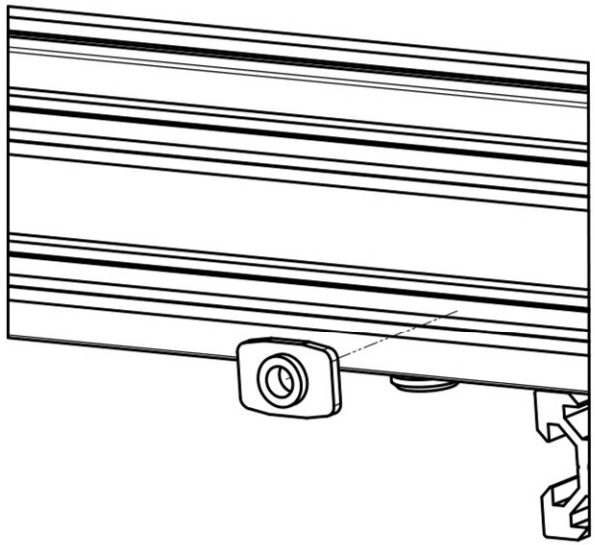
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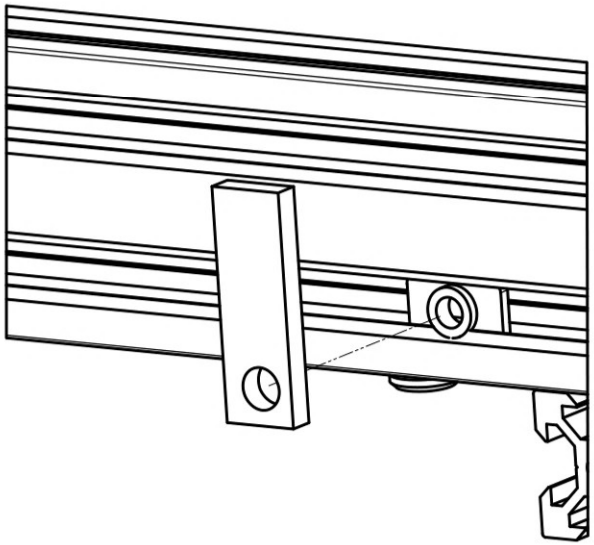
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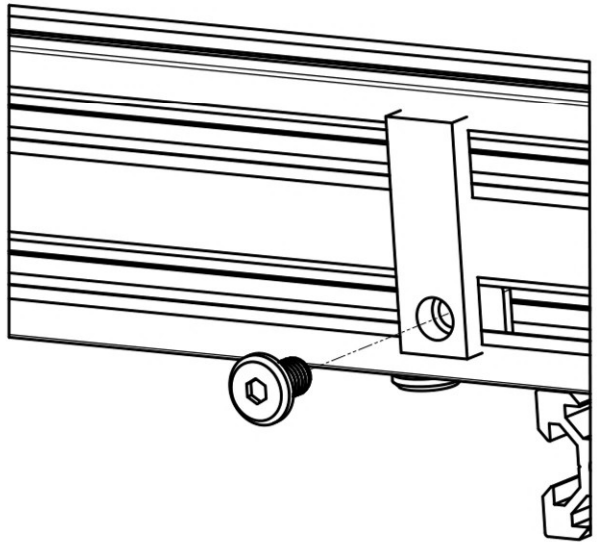
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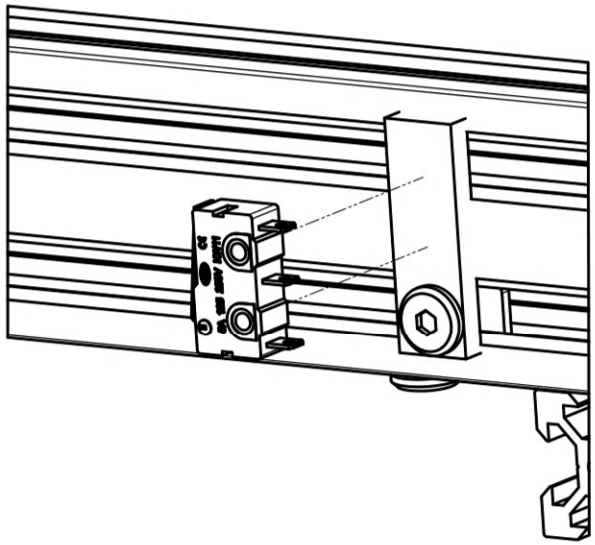
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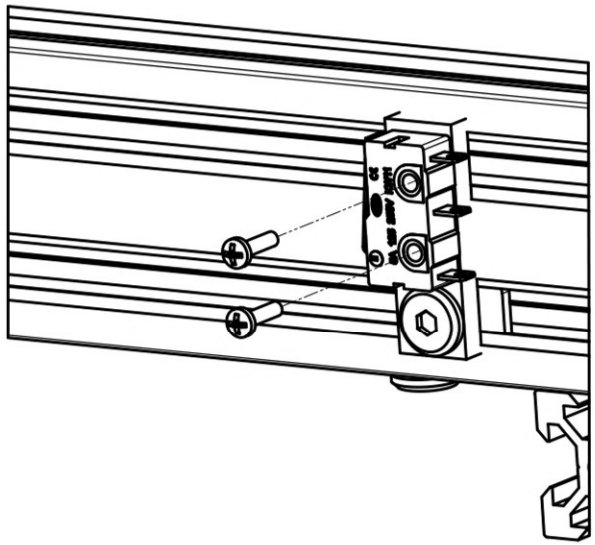
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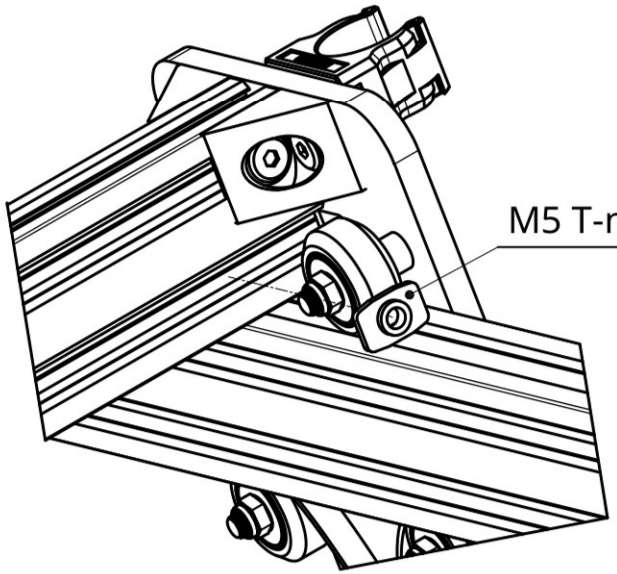
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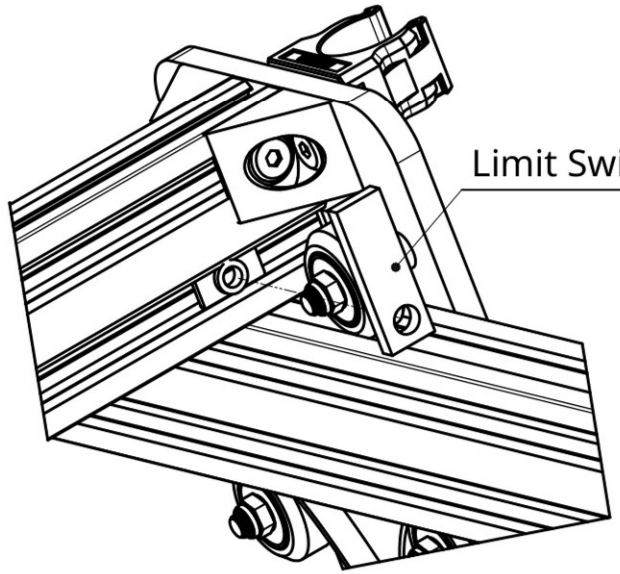
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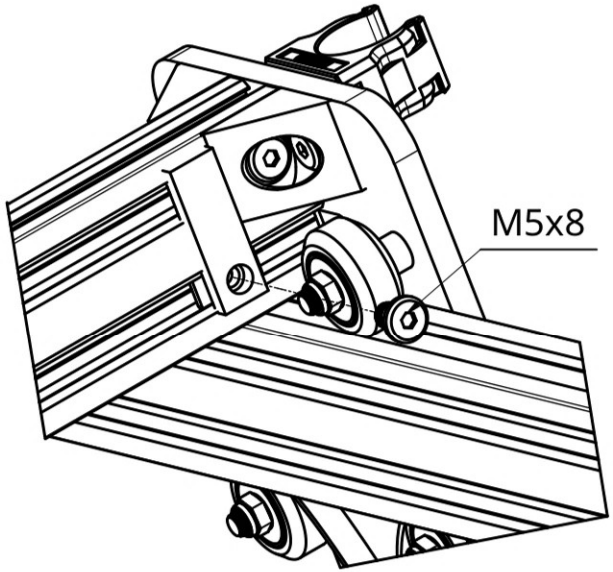
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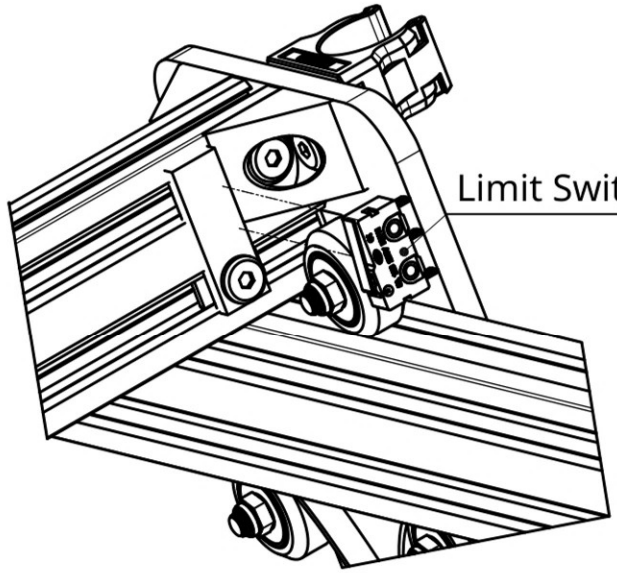
12



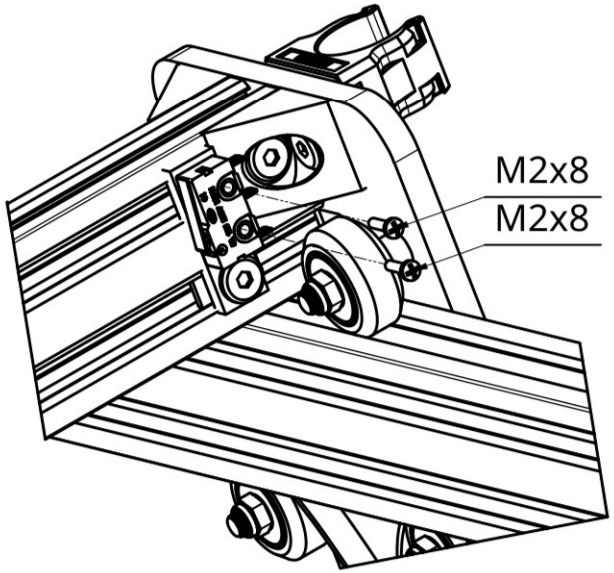
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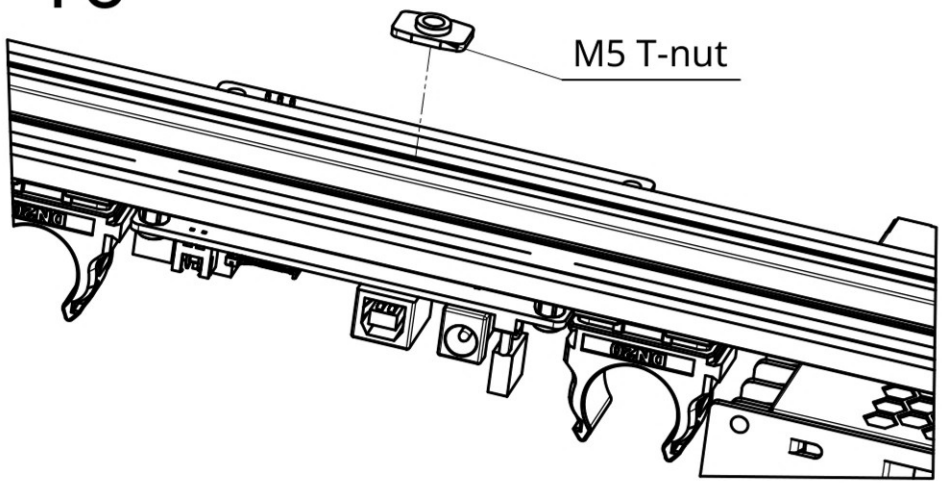
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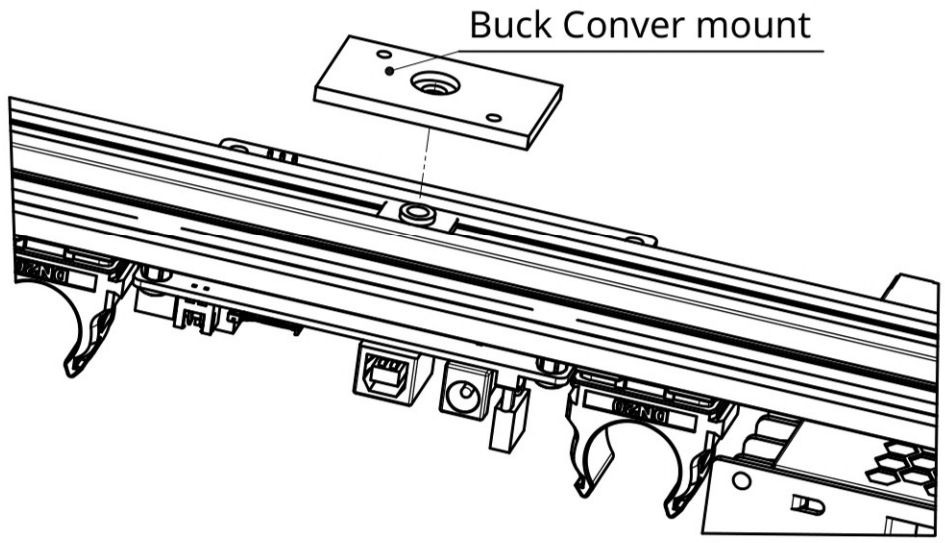
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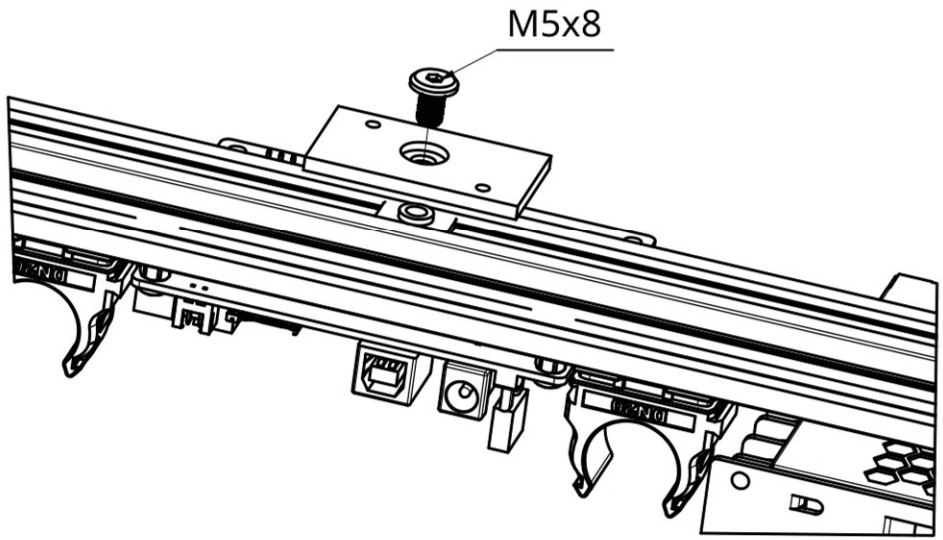
16



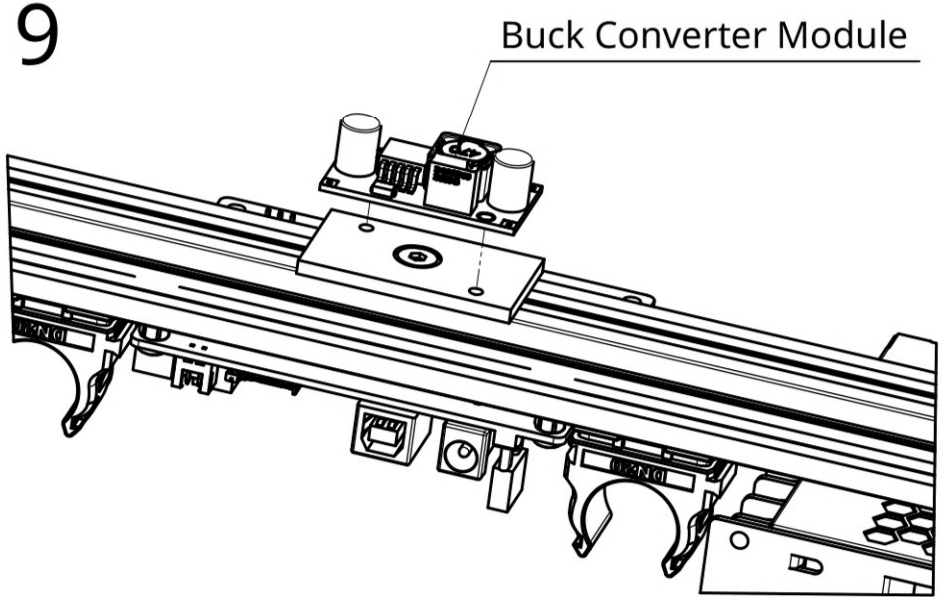
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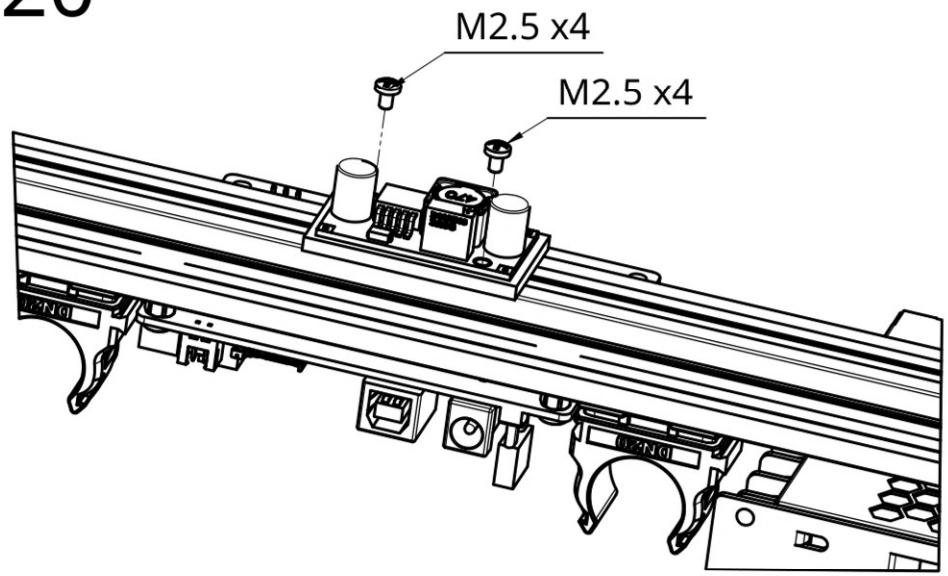
18



19

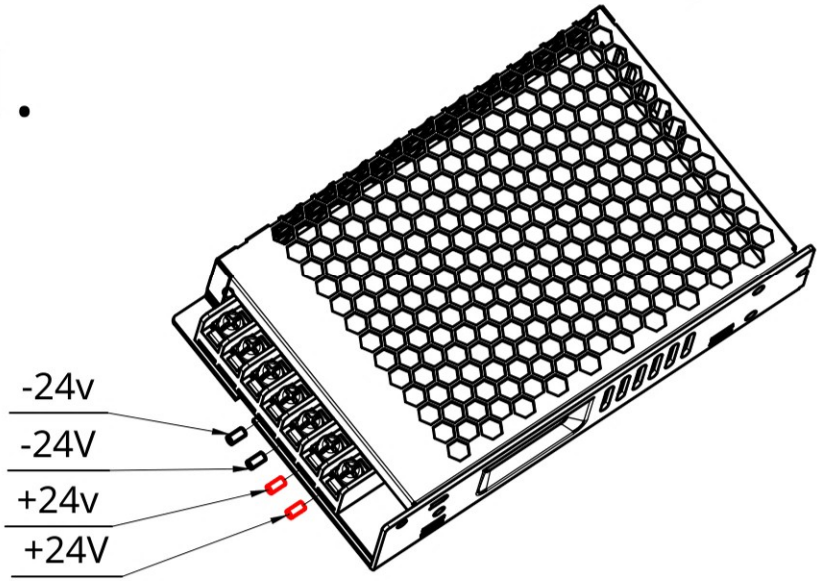


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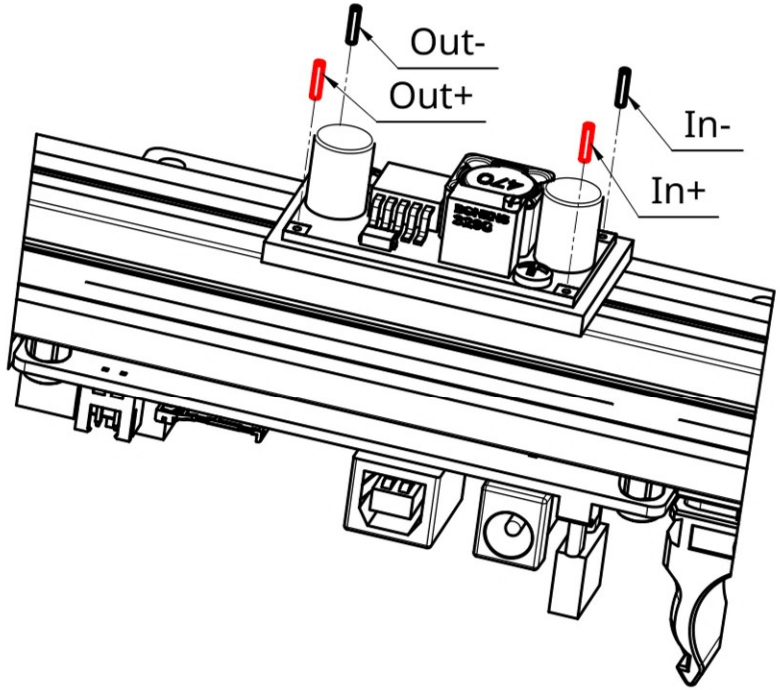


Wiring Assembly

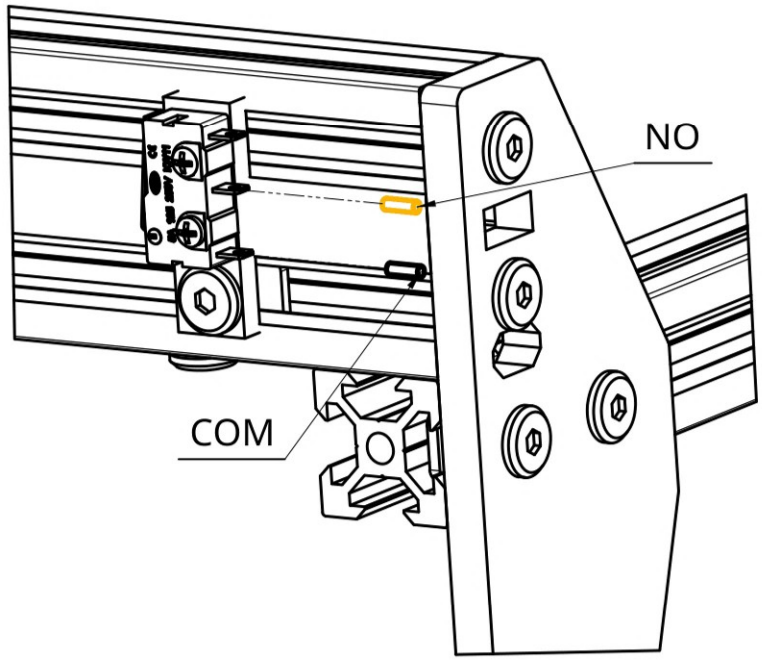
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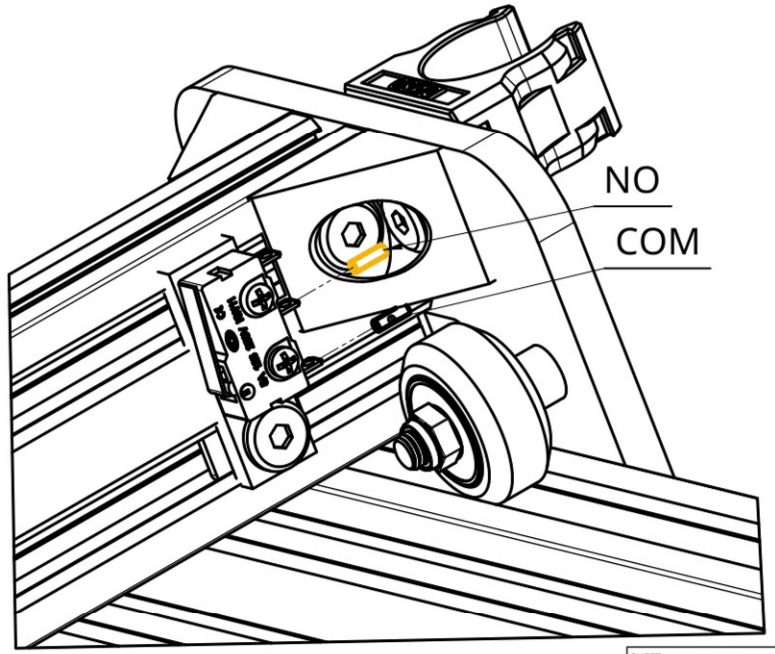
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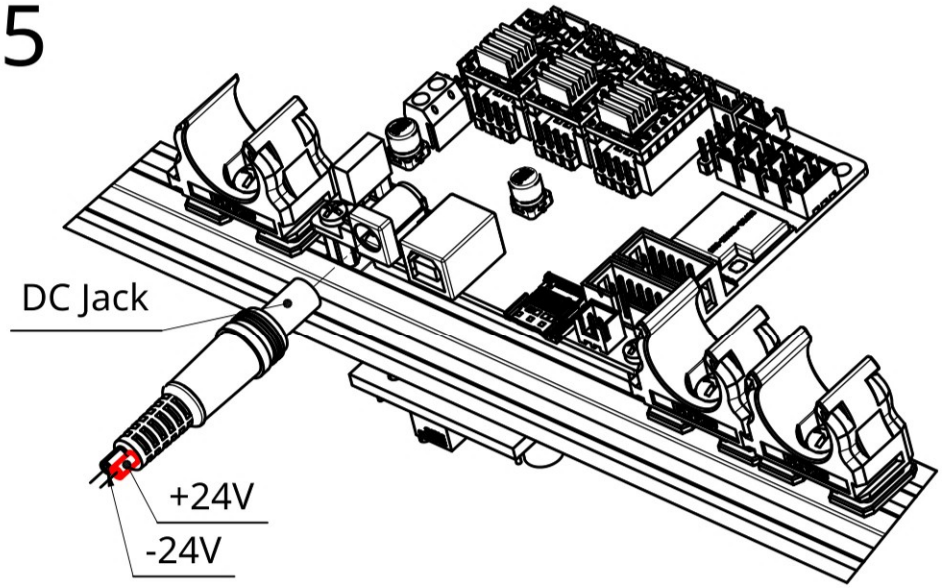
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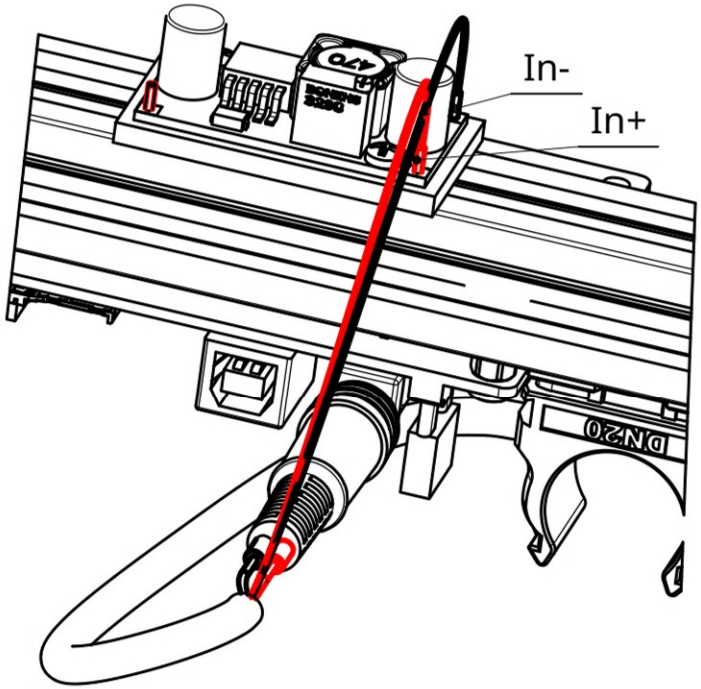
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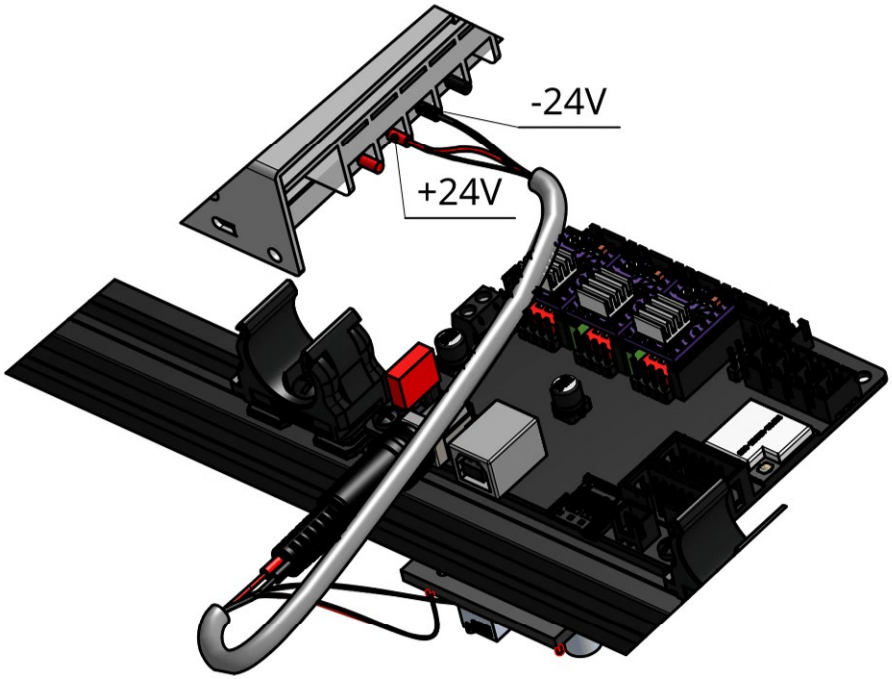
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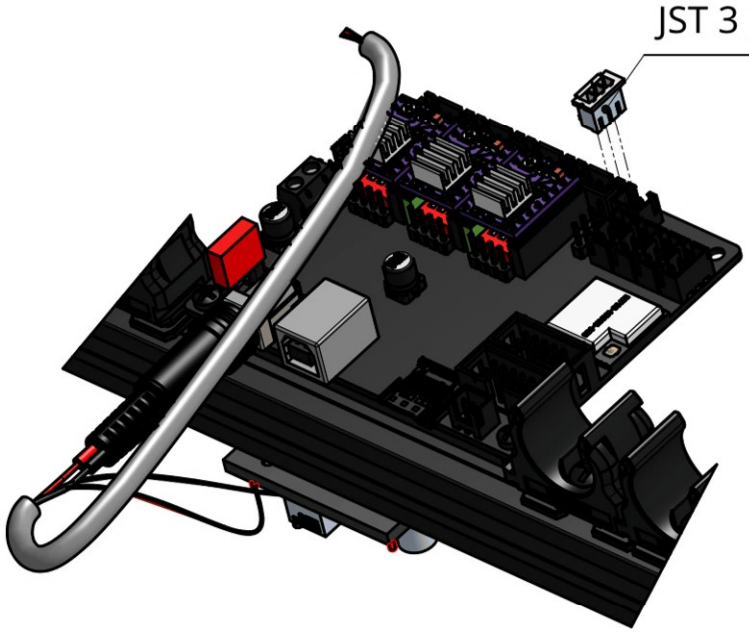
6



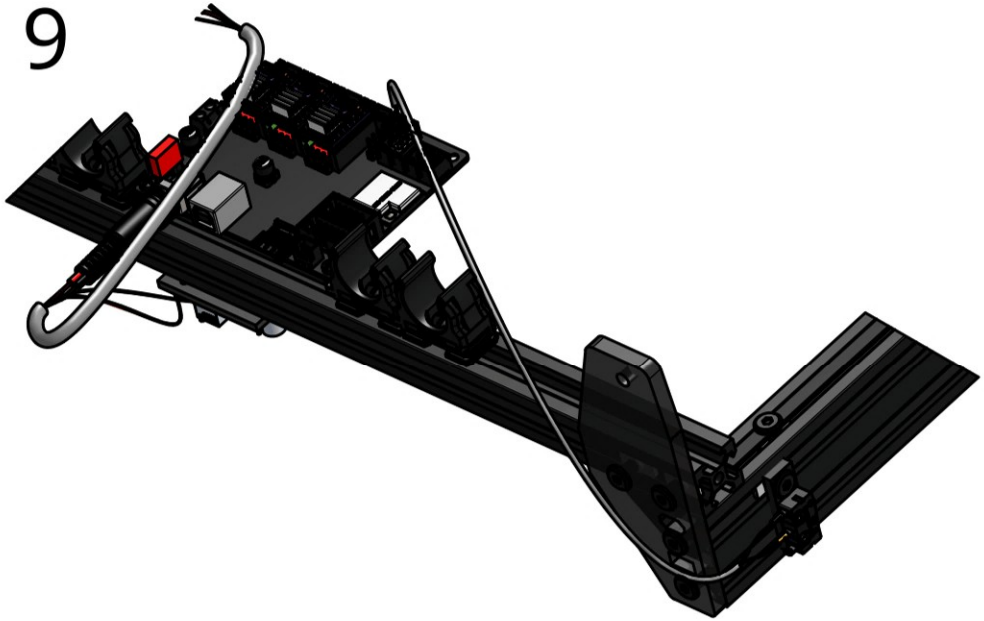
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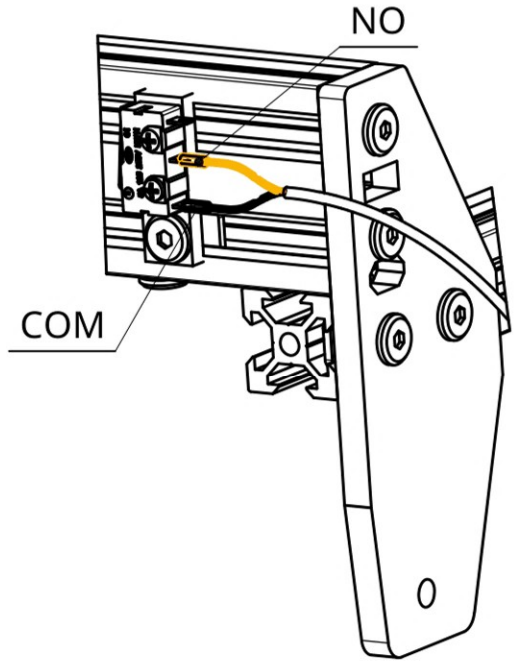
8



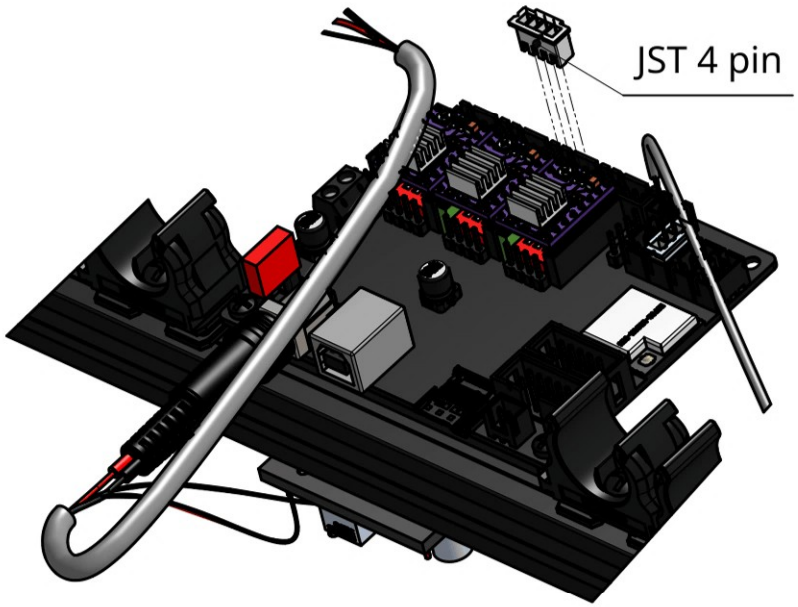
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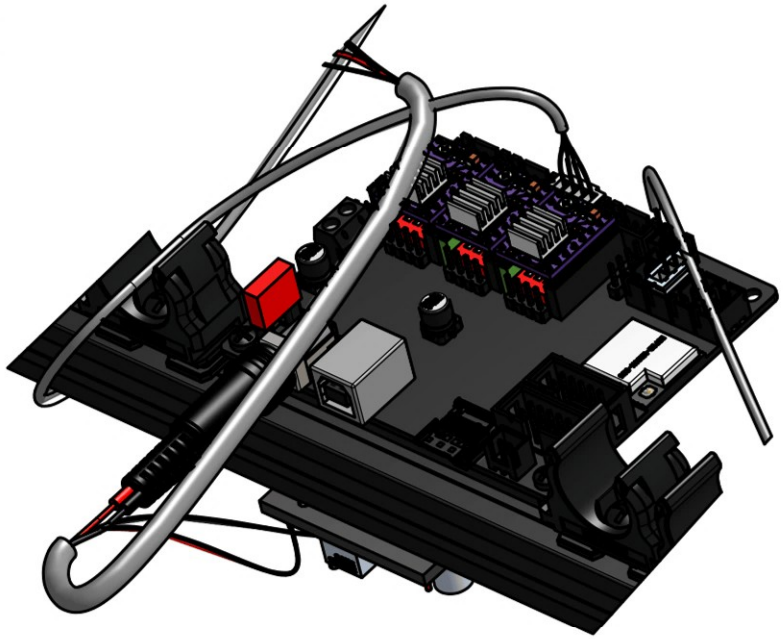
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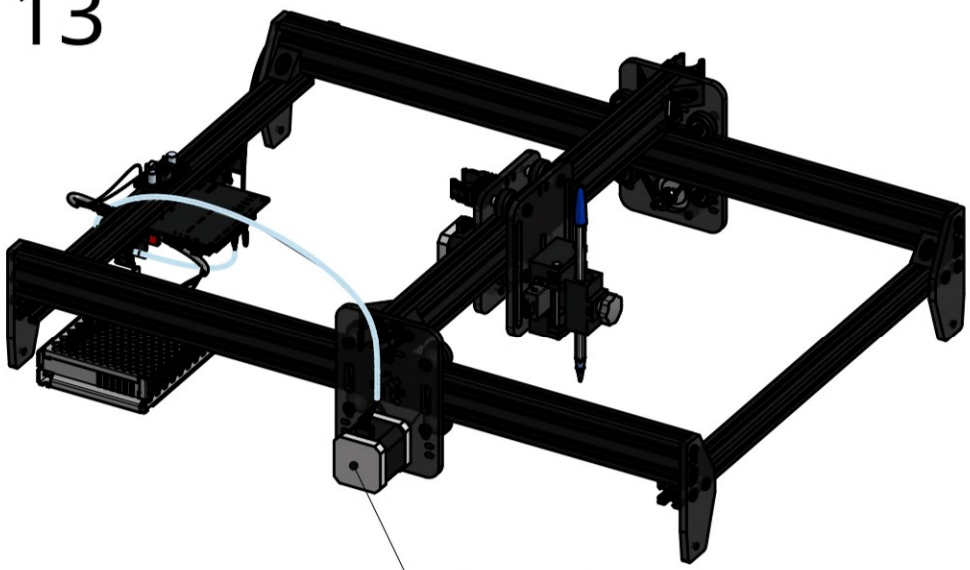
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12

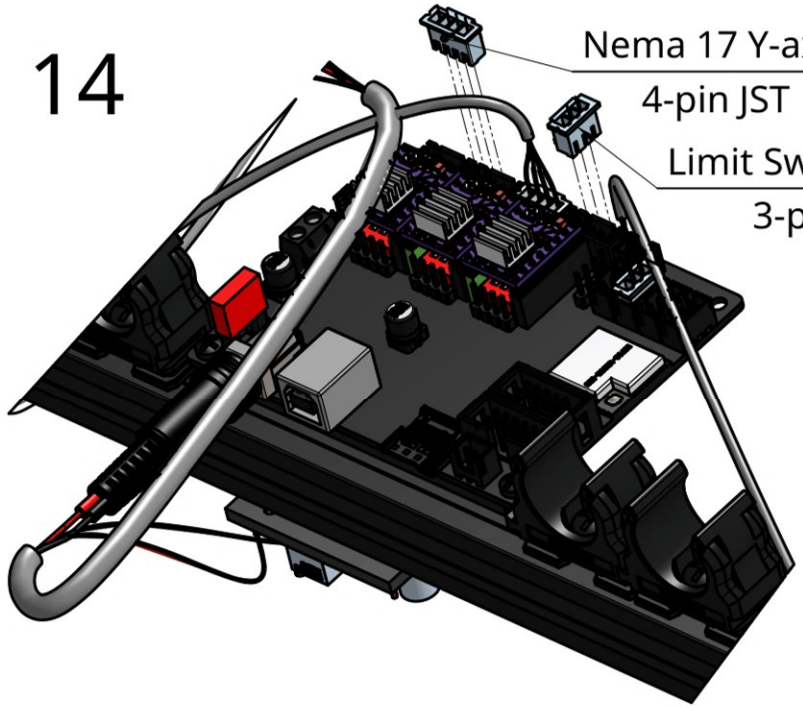


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Y-axis Nema 17 stepper motor

14



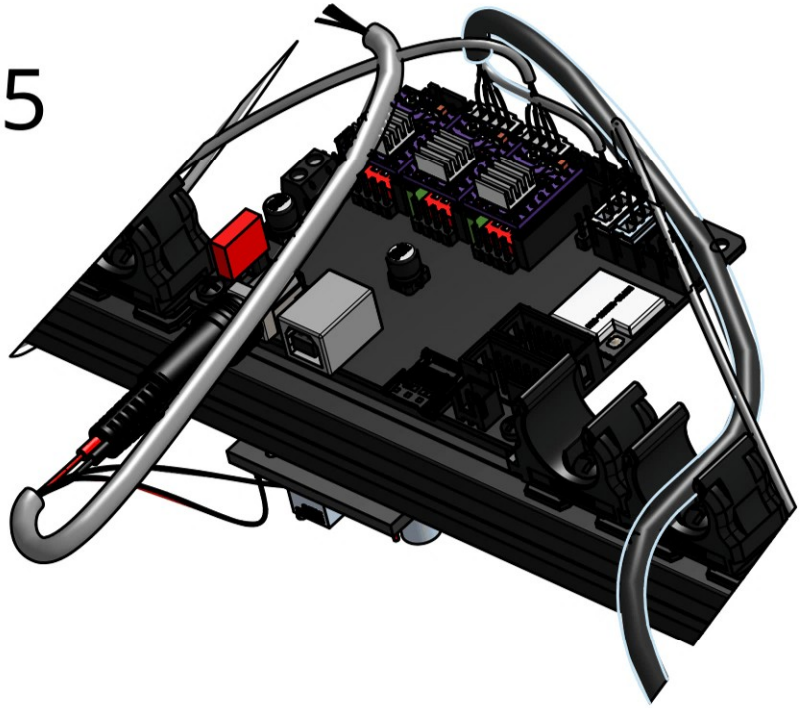
Nema 17 Y-axis

4-pin JST

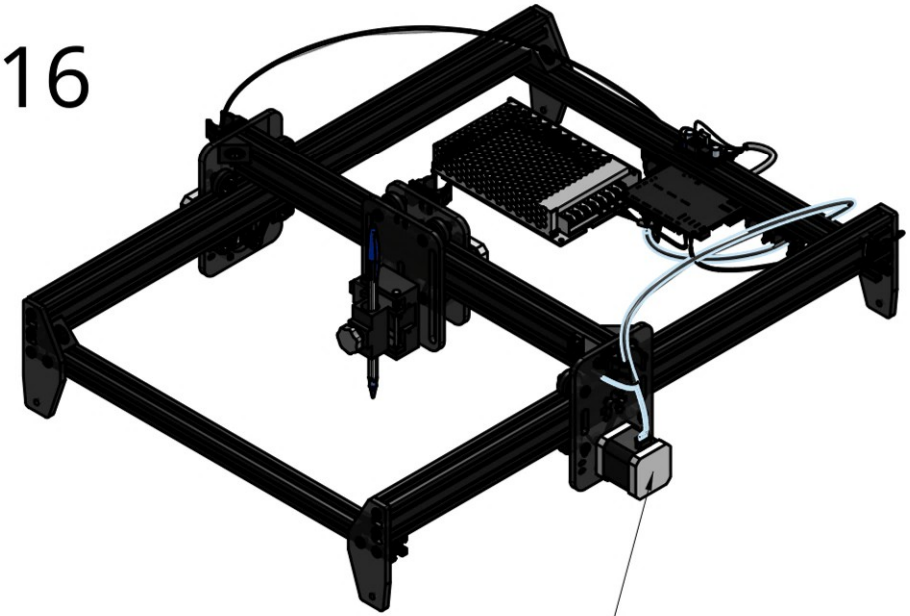
Limit Switch X-axis

3-pin JST

15

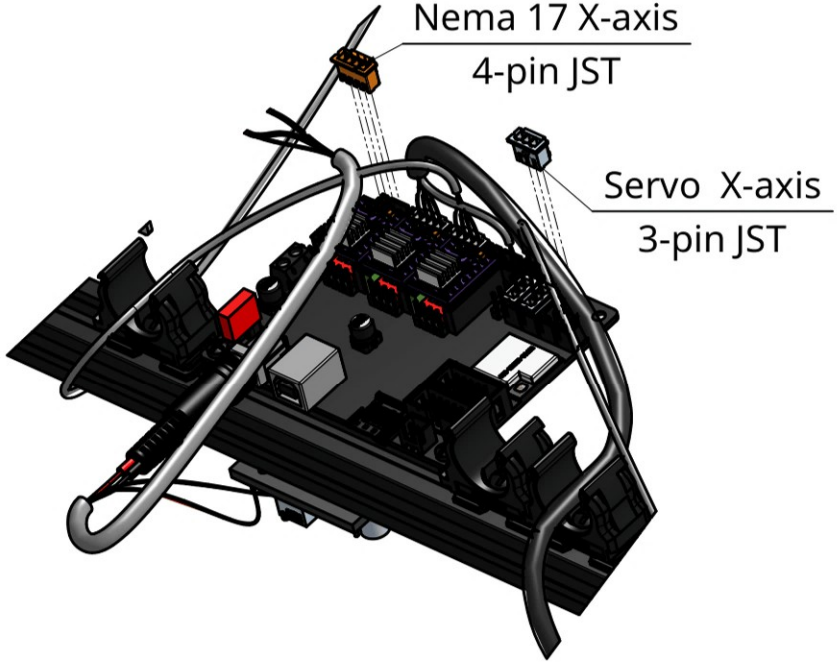


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Y-axis Nema 17 stepper motor

15



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17

